Continuous Change Detection of Flood Extents With Multisource Heterogeneous Satellite Image Time Series

Zhihao Wang, *Student Member, IEEE*, Xueqian Wang, *Member, IEEE*, Wei Wu, and Gang Li, *Senior Member, IEEE*

***Abstract*— Flood monitoring is of crucial importance for protecting lives and properties. Change detection (CD) methods on multisource remote sensing images have been widely used for flood extent monitoring. In this article, we propose a spatiotemporal fusion CD (STFCD) algorithm, exploiting the spatial dependence and temporal interaction of multisource heterogeneous (MSH) satellite image time series (SITS), to realize improved flood CD performance in comparison with existing methods. The proposed STFCD algorithm mainly contains two steps, i.e., spatial clustering and temporal fusion (TF). In the spatial clustering step, we propose a sparse Markov random field (MRF)-based strategy to exploit contextually spatial features in each image of MSH-SITS, which provides a larger local receptive field than the commonly used MRF. In the TF step, the historical information of flood detection results is employed as constraints to effectively reduce the effects of terrain shadows in synthetic aperture radar (SAR) images and cloud shadows and topography shadows in optical images on flood CD results of existing methods in accordance with the temporal dependence among MSH-SITS. Experiments on real MSH-SITS (containing Gaofen-1, Gaofen-3, Gaofen-6, Sentinel-1, and Sentinel-2 satellite images) covering Chinese Amur and Huma Rivers show that the overall flood CD accuracy of our proposed STFCD algorithm is higher than the other commonly used algorithms for CD of flood extents and demonstrate the robustness of our proposed STFCD algorithm.**

***Index Terms*— Change detection (CD), flood, multisource remote sensing, spatiotemporal fusion (STF).**

I. Introduction

DRAMATIC climate change caused by rising greenhouse gas emissions is triggering frequent occurrences of floods [1], [2], which are one of the costliest natural disasters for people [3], [4], [5]. To cope with the threat of floods, the real-time monitoring of floods is of crucial importance. With the advancement and development of remote sensing technologies, the remote sensing data derived from Earth observation (EO) satellites have been widely applied for flood monitoring due to their large monitoring range and high spatiotemporal resolution, which reduces the loss of lives and properties [6].

Single-source time-series remote sensing data, including synthetic aperture radar (SAR) images or optical images, have been commonly used for the task of continuous monitoring of flood areas [7]. Optical remote sensing images with good quality can be obtained under fine illumination conditions. In [8], a normalized difference surface water index (NDSWI)-based algorithm is proposed to produce flood detection results derived from the optical images. However, due to the principle of passive imaging, optical satellites are difficult to achieve flood detection missions in cases of cloud occlusion and cloud shadows. SAR images are generated by the active imaging mode that is hardly affected by lighting conditions [9], [10], [11]. Many flood detection algorithms [12], [13], [14] applied backscatter thresholds on SAR images, due to the very low backscatter of water areas, i.e., the reflected energy by the smooth and open water surface is near-completely not received by the SAR sensors [4]. Nevertheless, the quality of flood detection using SAR images may be degraded by high clutter and sidelobe levels therein [15].

Recently, there is a great development in the context of change detection (CD) of flood extents based on heterogeneous remote sensing images to comprehensively exploit the superiorities of SAR images and optical remote sensing images and to achieve timely flood warnings. The homogeneous pixel transformation (HPT)-based CD algorithms [16], [17] have been developed to transform heterogeneous remote sensing data into a homogeneous feature space to obtain the changed areas, which is performed by calculating the distances between two image patches in the semantic feature space. In [18] and [19], unsupervised deep neural networks are proposed based on SAR and optical remote sensing images to obtain the CD results in the high-dimensional feature space. These aforementioned algorithms [16], [17], [18], [19] focus on bitemporal heterogeneous CD analysis without the exploitation of historical flood information and may show limited flood detection performance in continuous monitoring cases using satellite image time series (SITS). This is because the exploitation of historical flood information is of crucial importance to construct the temporal interaction in the CD

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Zhihao Wang, Xueqian Wang, and Gang Li are with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: gangli@tsinghua.edu.cn).

Wei Wu is with the National Disaster Reduction Center, Ministry of Emergency Management, Beijing 100124, China.

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analysis among three or more SITS [20], [21], [22], [23]. In [5], a multiscale framework is proposed for rapid change analysis of flood monitoring (RCAFM) based on SAR image time series (ITS) with medium spatial resolution and medium temporal resolution.

With the development of remote sensing satellite constellations, the increased temporal resolution of the multisource heterogeneous (MSH)-SITS [24] improves the accuracy of the continuous flood CD. But few flood CD algorithms simultaneously utilize MSH-SITS, i.e., dense SAR and optical ITS for flood CD. The key issues of flood CD in MSH-SITS are the exploitation of spatial correlation and temporal interaction therein. Regarding the exploitation of spatial correlation, in [25], an iterative feedback-based CD (IFBCD) algorithm based on Markov random field (MRF) has been proposed to optimize the difference maps among the multiple satellite images to obtain the CD results. In [26], a spatial attention-based neural network is proposed to acquire improved CD results among the satellite remote sensing images. For temporal interaction in MSH-SITS, the combination of normalized change index and ratio image is employed based on the permanent water body mask via optical images to access accurate flood maps from SAR ITS [27]. In [28], the synergistic use of three threshold-based flood detection algorithms on MSH-SITS leads to a significant improvement in flood detection accuracy in comparison with single-source-based methods. However, the exploitation of spatial dependence in [25] has a limited receptive field on MSH-SITS, which may lead to the misjudgment of pixels with limited contextual features. In addition, the algorithms in [27] and [28] mainly focus on the design of global thresholds to find flood areas and pay little attention to exploiting the temporal synergy in MSH-SITS, where the local flood detection results may be affected by the topography shadows and cloud shadows in MSH-SITS.

To address the aforementioned problems of existing methods [25], [26], [27], [28], we propose a novel spatiotemporal fusion CD (STFCD) algorithm for the flood monitoring mission, where an effective flood spatial feature extraction module and a temporal fusion (TF) strategy [29] are established to enhance the exploitation of spatial dependence and temporal interaction based on MSH-SITS in comparison with existing methods. Our STFCD algorithm mainly includes two steps, i.e., spatial clustering and temporal fusion. In the spatial clustering step, the hierarchical fuzzy C-means clustering (HFCM) algorithm [30] is introduced to generate the initial flood detection results. Then, we propose the sparse Markov random field (SMRF)-based operation to iteratively optimize the initial flood detection results with the dynamical conceptive fields than the second-order MRF [31]. In the temporal fusion step, we apply an efficient fusion strategy to exploit the temporal dependence of MSH-SITS to realize the cross-reduction of false alarms derived from terrain shadows in SAR images and cloud shadows and topography shadows in optical images, which further refines the spatial clustering results without a noticeable increase of computational burden. Then, we perform CD operations based on the final flood maps. The experimental results are performed on MSH-SITS covering Chinese Amur and Huma Rivers with five optical satellite images originated from three different optical satellite sensors and seven SAR images derived from two different SAR satellite sensors. The experimental results have verified that the proposed STFCD algorithm has superior CD performance of flood extents than the existing commonly used CD algorithms.

Part of this work has been presented in [32]. In this article, we further optimize both the SMRF energy function and dynamic flood thresholds updating the rules of each pixel to obtain better local flood features and refine the temporal fusion strategy to exploit the temporal relevancy of historical flood detection results in MSH-SITS with the larger test dataset and more comparison methods. This article also provides more detailed method analyses than [32].

This article is organized as follows. Section I introduces the study background and related works. Section II describes the studied area and dataset. Section III propounds the structure of flood monitoring framework and elaborates on the proposed STFCD algorithm. Section IV provides and discusses the experimental results based on MSH-SITS derived from Gaofen-1 (GF-1) satellite [33], Gaofen-3 (GF-3) satellite [34], Gaofen-6 (GF-6) satellite [35], Sentinel-1 (S-1) satellite, Sentinel-2 (S-2) satellite [36], [37], and aeronautical reconnaissance coverage geographic information system (ArcGIS) imagery [38], [39]. Section V presents the findings of the research and prospects the future work.

II. Case Study and Dataset

A. Study Area

The main area studied in this article is the flood hazard region in Huma Town (HT), Heilongjiang Province, China, from May to June 2021. HT is located at the intersection of the Huma River and the Amur River. The Amur River, i.e., Heilongjiang [40], is the eighth-largest river in the world and the most important international river in East Asia, flowing through Mongolia, Russia, China, and North Korea [41]. The Huma River is one of the biggest upper main tributaries of the Amur River and is also the largest tributary of the Amur River flowing through HT.

According to the observation data of the Heilongjiang Province Hydrology and Water Resources Center (HPHWRC), Harbin, in China, since June 1, 2021, 33 stations of nine rivers including the mainstream of the Amur River and its tributaries have had floods exceeding the warning level of 0.03–4.2 m [42]. In late June 2021, the greatest flood in HT since 1983 [43] caused damages to 266.40 square kilometers of farmlands and 756 houses [44]. On June 30, the water level in the section above HT retreated below the warning level [43]. The flood extent in this study is between 51° 33′ 57.66″ N and 51° 46′ 25.67″ N and 126° 31′ 35.62″ E and 126° 50′ 26.94″ E (see Fig. 1). The top-left/middle/right images in Fig. 1 indicate that the actual process of flood evolution is from the upstream to the downstream of the Amur River according to the time of flood peak measured by each hydrological station (HS). Since the studied area is between the Huma HS and Sandaoqia HS, it can be inferred that the flood peak of HT reached around June 27, 2021. It is worth emphasizing that the application of our method is not limited to the HT area. Here, we use

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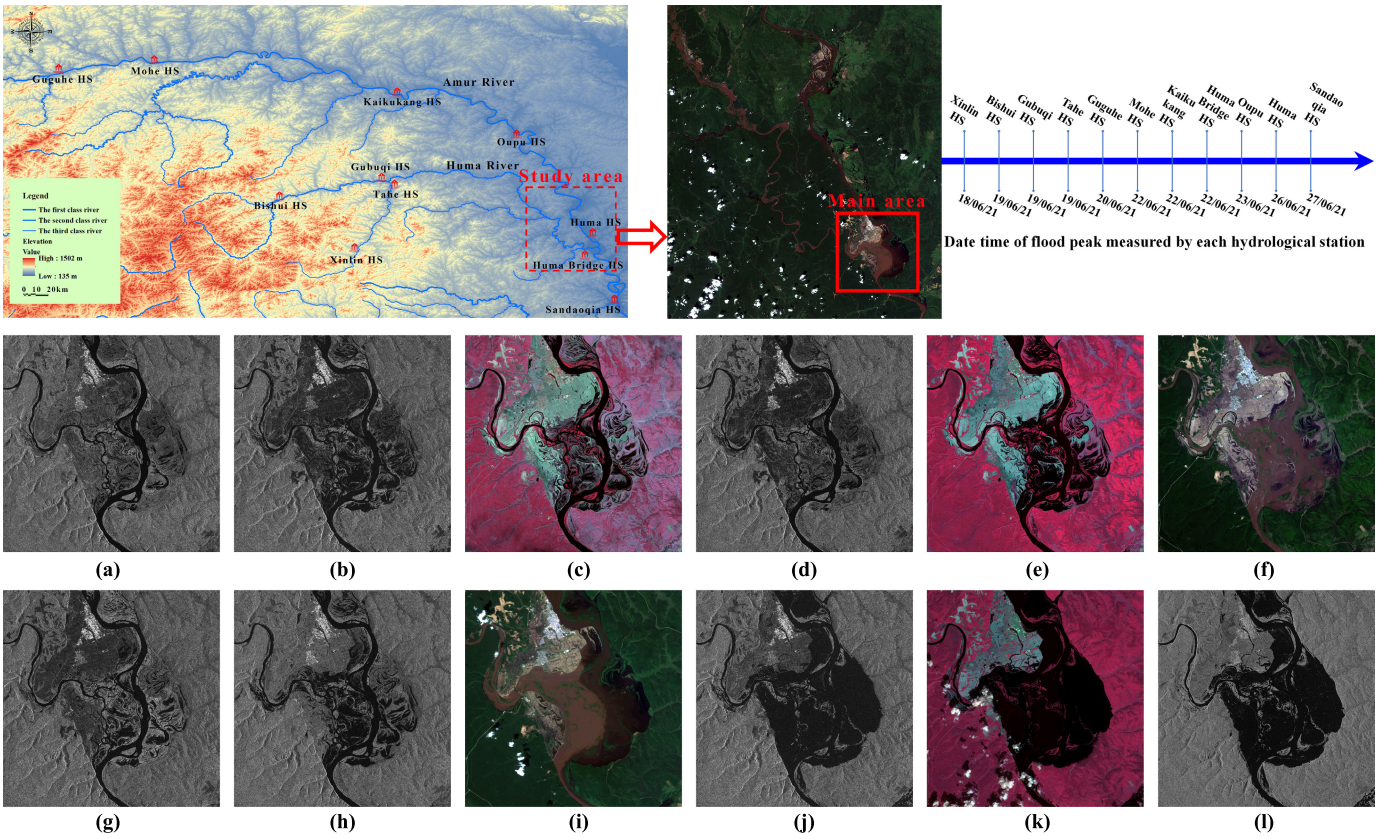


Fig. 1. Study area and thumbnails of MSH-SITS, 2021. The top-left image is the DEM, ©ASTER Global DEM V3 [46], and HS referenced in this work; the top-middle image is the location of the studied area, i.e., GF-6 data obtained on June 21, 2021, ©NRSRSCSP; and the top-right image is the date time of flood peak measured by each HS [42], [43]: (a) S-1 data (Ψ_1) acquired on May 14, 2021, ©ESA; (b) S-1 data (Ψ_2) collected on May 19, 2021, ©ESA; (c) S-2 data (Ψ_3) captured on May 21, 2021, ©ESA; (d) S-1 data (Ψ_4) accessed on May 26, 2021, ©ESA; (e) S-2 data (Ψ_5) attained on June 7, 2021, ©ESA; (f) GF-1 data (Ψ_6) gathered on June 8, 2021, ©NRSRSCSP; (g) S-1 data (Ψ_7) received on June 12, 2021, ©ESA; (h) S-1 data (Ψ_8) obtained on June 19, 2021, ©ESA; (i) GF-6 data (Ψ_9) acquired on June 21, 2021, ©NRSRSCSP; (j) S-1 data (Ψ_{10}) collected on June 24, 2021, ©ESA; (k) S-2 data (Ψ_{11}) captured on June 27, 2021, ©ESA; and (l) GF-3 data (Ψ_{12}) acquired on June 27, 2021, ©NRSRSCSP.

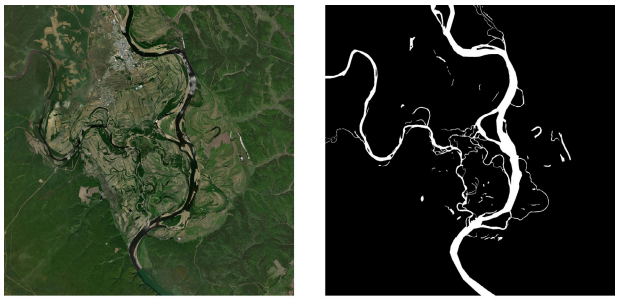


Fig. 2. ORGB and GT of the river system in HT before the flood event: (a) original optical remote sensing image with the RGB bands, ©ArcGIS imagery and (b) GT of ORGB manually labeled by the ENVI software (black: background and white: water).

TABLE I
BASIC PARAMETERS OF ORIGINAL MSH-SITS COVERING HT

Original MSH-SITS	Swath	Dimension	Pixle size	Spatial resolution
GF-1	200 km	4 bands	$12k \times 13k$	16 $m \times 16$ m
GF-3	100 km	1 band	$36k \times 22k$	10 $m \times 10$ m
GF-6	$\geq$800 km	8 bands	$50k \times 15k$	16 $m \times 16$ m
S-1	250 km	2 bands	$25k \times 16k$	3.5 $m \times 22$ m
S-2	290 km	13 bands	$11k \times 11k$	20 $m \times 20$ m

the abundant satellite data of the HT area to verify the effectiveness of our method.

B. Dataset

We use GF-1, GF-3, GF-6, S-1, and S-2 corresponding to HT as our MSH-SITS data and refer to ArcGIS imagery data at HT as the contrast image. ArcGIS imagery data were collected before May 2021, and the rest of the data were collected from May 14 to June 27, 2021. In this study, an original optical remote sensing image of RGB bands (ORGB) with a resolution of 1 m corresponding to HT was acquired from ArcGIS imagery. Then, we referred to the water system of HT labeled in Bing map and ORGB of HT [see Fig. 2(a)] and labeled the original watershed to obtain the ground truth (GT) of the water system in the study area before the flood [see Fig. 2(b)]. We use the environment for visualizing images (ENVI) software to achieve the annotation process in this study.

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TABLE II
DETAILS OF MSH-SITS IN THE MAIN STUDY AREA

Observation date	Thumbnail	Sensor type	Revisit period	Spatial resolution
Before 2021/05	Fig. 2 (a)	ArcGIS (optical)	–	$1\ m \times 1\ m$
2021/05/14 21:44:40	Fig. 1 (a)	S-1 (SAR)	6 days	$3.5\ m \times 22\ m$
2021/05/19 21:52:50	Fig. 1 (b)	S-1 (SAR)	6 days	$3.5\ m \times 22\ m$
2021/05/21 02:45:51	Fig. 1 (c)	S-2 (optical)	5 days	$20\ m \times 20\ m$
2021/05/26 21:44:16	Fig. 1 (d)	S-1 (SAR)	6 days	$3.5m\ \times 22\ m$
2021/06/07 02:35:51	Fig. 1 (e)	S-2 (optical)	5 days	$20\ m \times 20\ m$
2021/06/08 12:06:38	Fig. 1 (f)	GF-1 (optical)	4 days	$16\ m \times 16\ m$
2021/06/12 21:52:01	Fig. 1 (g)	S-1 (SAR)	6 days	$3.5\ m \times 22\ m$
2021/06/19 21:44:18	Fig. 1 (h)	S-1 (SAR)	6 days	$3.5\ m \times 22\ m$
2021/06/21 02:51:05	Fig. 1 (i)	GF-6 (optical)	4 days	$16\ m \times 16\ m$
2021/06/24 21:52:27	Fig. 1 (j)	S-1 (SAR)	6 days	$3.5\ m \times 22\ m$
2021/06/27 02:35:51	Fig. 1 (k)	S-2 (optical)	6 days	$20\ m \times 20\ m$
2021/06/27 09:37:50	Fig. 1 (l)	GF-3 (SAR)	3/1.5 days	$10\ m \times 10\ m$

According to Natural Resources Satellite Remote Sensing Cloud Service Platform (NRSRSCSP) and European Space Agency (ESA), basic parameters, including swath, dimension, pixel size, and spatial resolution of original MSH-SITS employed in this study, are listed in Table I. Specifically, we select the near-infrared band (NIRB) image of GF1 and GF6, the single polarimetric SAR image of S-1 and GF-3, and B08, B11, and B12 bands of S-2 as original input images. Details of preprocessed MSH-SITS data at 13 time points in HT are listed in Table II. Since we focus on the CD of large-scale flood extents near residential areas and the limited overlapping area of MSH-SITS, we employ the data preprocessing method (in Section III-B) to intercept the main flood hazard regions (with the pixel size of 1401 × 1401) in MSH-SITS data as reported in [42] and [43].

III. PROPOSED METHOD

A. Overview of the Proposed STFCD

The framework of our proposed STFCD algorithm is shown in Fig. 3, including data preprocessing, spatial clustering, and temporal fusion stages. Compared with existing commonly used methods, the main innovations of our STFCD algorithm are the SMRF-based flood features with better spatial dependence and the utilization of historical flood detection results to reduce the effects of terrain/cloud shadows on the flood CD results. We use $\mathbf{\Psi} = \{\mathbf{\Psi}_t\}_{1 \le t \le S}$ to represent S images in MSH-SITS, where $\mathbf{\Psi}_t$ indicates the image obtained at the time point t.

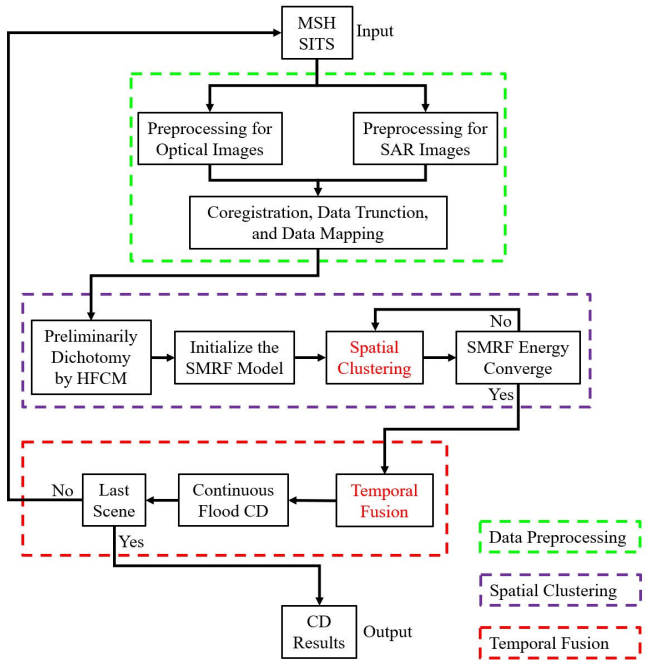


Fig. 3. Framework of our STFCD algorithm for CD of flood extents.

In the data preprocessing stage, radiometric correction, atmospheric correction, and orthorectification for optical images, and radiometric correction, geocoding, and probabilistic patch-based (PPB) weight filtering [45] for SAR images, image coregistration, data truncation, and data mapping are performed to enhance the quality of MSH-SITS and make it convenient to extract flood features. The pixel size of each image in $\mathbf{\Psi}$ after data preprocessing is $\mathcal{R}^{M \times N}$.

In the spatial clustering stage, we preliminarily obtain the flood detection results of $\mathbf{\Psi}$. First, the initial (binary) flood detection results of each image in MSH-SITS are obtained based on the HFCM algorithm [30]. Then, we iteratively optimize the SMRF energy function to enhance the boundaries of initial flood detection results in accordance with contextually spatial features in MSH-SITS. The energy-adaptive sparse expansion of the spatial neighborhood system (SNS) in SMRF adaptively increases the receptive fields for flood detection without a noticeable increase in computation cost compared with the commonly used second-order SNS in MRF [31].

In the temporal fusion stage, we alleviate the effects of cluttered background noise and finally generate the flood CD results of $\mathbf{\Psi}$. First, we initialize the temporally fused constraints based on the spatial clustering results and the global flood area proportion of $\mathbf{\Psi}$. Then, the spatiotemporally fused flood detection results are obtained by reducing the suspected flood areas that are background noise in practice. This temporal fusion step utilizes the historical information of the recent flood detection results as clustering constraints to realize the global optimization of the current flood detection results based on temporal interaction characteristics in MSH-SITS. Finally, we perform the matrix difference operation on the final flood features extracted from MSH-SITS to obtain the flood CD results between the specific scenes.

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B. Data Preprocessing

The preprocessing steps of the wide field of view (WFV) data from GF-1 and GF-6 include radiometric correction, atmospheric correction, and orthorectification. We use the ENVI software to achieve the aforementioned preprocessing steps. The preprocessing steps of hyperfine band3 level 1A (FSII L1A) data from GF-3 include radiometric calibration and geocoding. Moreover, we use the Sentinel Application Platform (SNAP) software to preprocess the S-1, GF-3, and S-2 data. Basically, we apply the ASTER Global digital elevation model (DEM) V3 [46] to achieve geometric correction for S-1 and GF-3 data and convert the S-2 data with B8, B12, and B11 bands to the gray-scale images with a resolution of 20 × 20 m. Then, all the SAR images are preprocessed by the PPB weight filter [45] to reduce the speckle noise in SAR images.

After these preprocessing steps of each image, all the images in MSH-SITS are coregistered by the ENVI software with the same spatial resolution of 16 × 16 m. Specifically, we set Fig. 2(a) as the primary image, and use the image registration workflow in ENVI to register Fig. 1(a)–(l) with the matching algorithm, minimum matching score, geometric model, transform, maximum allowable error per tie point, requested number of tie points, search window size, matching window size, and interest operator setting as cross correlation/mutual informations (for Fig. 1(c), (e), (f), (i), and (k) and Fig. 1(a), (b), (d), (g), (h), (j), and (l), respectively), 0.60, fitting global transform, first-order polynomial, 2.00, 24, 255, 61, and "Forstner," respectively. In the generated ground control points (GCPs) lists, we manually replace the GCPs with significant errors. Moreover, we truncate and map the coregistered MSH-SITS into the same intensity range $[V_{\min}, V_{\max}]$ to eliminate outliers (e.g., $V_{\min} = 0$ and $V_{\max} = 255$), where $V_{\min}$ and $V_{\max}$ mean the minimum and maximum values of pixel intensity in MSH-SITS, respectively. The truncation and mapping operations alleviate the problem of inconsistent data distributions in MSH-SITS [47]. The pixel intensities of the water areas are generally low according to the spectral characteristics of water in NIRB [48] and the near-completely reflection of radar waves on the water surface [4]. Considering these physical properties of water, we only use optical images with NIRBs and SAR images in MSH-SITS as the initial input after all the data preprocessing steps, which are generally available in flood monitoring tasks [4], [28].

C. Spatial Clustering

Considering the value of neighborhood pixels, the improved MRF in [49] extracts the feature information of neighboring pixels to reduce the over-smoothing problem caused by fixed spatial energy function and the edge-aware MRF in [50] takes the first-order and second-order SNSs into account, resulting in refined change edges. The iterative process of the spatial clustering step is summarized in Algorithm 1. First, we initialize the pixel intensity threshold matrix using the HFCM [30] algorithm to generate the initial flood detection results. Then, the flood detection results are iteratively updated at each iteration based on the dynamical threshold matrix of pixels to refine flood detection results by minimizing the SMRF energy function. The iteration is terminated when the decay ratio of the SMRF energy is smaller than 0 or the maximum of the iteration index is reached.

Algorithm 1 Spatial Clustering Step

Input: MSH-SITS $\boldsymbol{\Psi} = \{\boldsymbol{\Psi}_t\}_{1\le t\le S}$.

Initialization: Let $iter$ represent the iteration index and let $iter = 0$. For $t = 1, \ldots, S$, the initialized dynamical threshold matrix is $\boldsymbol{Th}_t^{iter} \in \mathcal{R}^{M\times N}$ in accordance with HFCM algorithm [see (1)]. The spatial clustering results are initialized as $\boldsymbol{D}_t^{iter} \in \mathcal{R}^{M\times N}$, where if $\boldsymbol{\Psi}_t(i,j) < \boldsymbol{Th}_t^{iter}(i,j)$, $\boldsymbol{D}_t^{iter}(i,j) = 0$, otherwise, $\boldsymbol{D}_t^{iter}(i,j) = 1$, for $t = 1, \ldots, S$, $i = 1, 2, \ldots, M$, and $j = 1, 2, \ldots, N$.

repeat

  Let $iter = iter + 1$ and calculate the SMRF energy function $U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{iter-1}\right)$ [see (2)].

  Update $\boldsymbol{Th}_t^{iter}$ [see (10)-(11)] by applying the spatial constraints $\boldsymbol{d}_t^{iter}$. Then, update the iterative spatial clustering results $\boldsymbol{D}_t^{iter} \in \mathcal{R}^{M\times N}$ based on $\boldsymbol{Th}_t^{iter}$ when all the pixels satisfied with $\{(i,j) \mid$ if $\boldsymbol{D}_t^{iter}(i,j) = 0$ and $\boldsymbol{Dis}_t^{iter}(i,j) < \varepsilon_3\}$ are traversed.

  Calculate $U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{iter}\right)$ [see (2)] and decay ratio EDR_t between $U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{iter}\right)$ and $U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{iter-1}\right)$ [see (3)]. Then, update the SNS in SMRF [see (4)] depending on EDR_t. (4)]

until; $\left(U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{iter-1}\right) - U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{iter}\right)\right) < 0$ or $iter = itermax = 100$;

Output: The final spatial clustering results $\boldsymbol{D}_t = 1 - \boldsymbol{D}_t^{iter-1}$

Flood pixels are often with low intensities in NIRB and SAR images [4], [48]. First, we initialize the spatial clustering results $\boldsymbol{D}_t^{\text{iter}} \in \mathcal{R}^{M\times N}$, where if $\boldsymbol{\Psi}_t(i,j) < \boldsymbol{Th}_t^{\text{iter}}(i,j)$, $\boldsymbol{D}_t^{\text{iter}}(i,j) = 0$; otherwise, $\boldsymbol{D}_t^{\text{iter}}(i,j) = 1$. $\boldsymbol{Th}_t^{\text{iter}}$ is the decision threshold matrix, iter denotes the iteration index, and iter is initialized as 0, for $t = 1, \ldots, S$, $i = 1, 2, \ldots, M$, and $j = 1, 2, \ldots, N$. To determine $\boldsymbol{Th}_t^{\text{iter}}$, we separate $\boldsymbol{\Psi}_t$ into several intensity intervals by employing the HFCM [30] algorithm, which has been used to distinguish flood and nonflood areas in MSH remote sensing data [30]. Let $\boldsymbol{C}_t = \left[C_t^1, C_t^2, \ldots, C_t^8\right]$ denote eight clustering centers (in the ascending order) of $\boldsymbol{\Psi}_t$ derived from the HFCM [30] algorithm. For iter = 0, $\boldsymbol{Th}_t^{\text{iter}}$ is initialized as

$$\boldsymbol{Th}_t^0 = \begin{cases} \varepsilon_1, & \text{if } \left(C_t^1 + C_t^2\right)/2 < \varepsilon_1 \\ (C_t^1 + C_t^2)/2, & \text{if } \varepsilon_1 \le (C_t^1 + C_t^2)/2 < \varepsilon_2 \\ \varepsilon_2, & \text{otherwise} \end{cases} \tag{1}$$

where $\varepsilon_1 > 0$ denotes the initially reliable threshold of flood detection and $\varepsilon_2 > \varepsilon_1 > 0$ indicates a temporary threshold of flood detection. As shown in Fig. 4, when the pixel intensity satisfies $\boldsymbol{\Psi}_t(i,j) < \varepsilon_1$, the current pixel is preclassified as a flood pixel. When the pixel value satisfies $\boldsymbol{\Psi}_t(i,j) > \varepsilon_2$, the current pixel belongs to the nonflood area. Otherwise, pixels with $\varepsilon_1 \le \boldsymbol{\Psi}_t(i,j) \le \varepsilon_2$ represent the suspected flood areas (see Fig. 4) and we will iteratively minimize the SMRF energy function [see (2)] to search the reliable threshold for each suspected pixel. In (1), $(C_t^1 + C_t^2)/2$ represents the average intensity of the pixels with relatively low intensities in the

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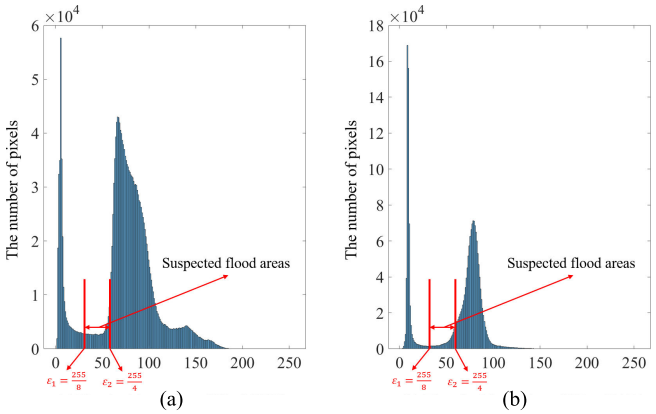


Fig. 4. Initialization settings of flood detection thresholds. Take NIRB [Ψ_5, i.e., Fig. 1(e)] and SAR [Ψ_{12}, i.e., Fig. 1(l)] images of the pixel intensity in the range of [0, 255] after truncation and mapping as an example: (a) pixel intensity of Ψ_5 (NIRB) and (b) pixel intensity of Ψ_{12} SAR.

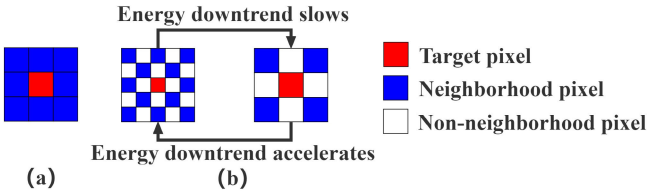


Fig. 5. Interpretation of the SNS: (a) SNS in MRF [31] and (b) SNS in our SMRF.

current image. If $(C_t^1 + C_t^2)/2$ is smaller than ε_1, then we set $\boldsymbol{Th}_t^0 = \varepsilon_1$ to obtain most of flood pixels. If $(C_t^1 + C_t^2)/2$ is larger than ε_2, the current image may contain fewer flood pixels and we set $\boldsymbol{Th}_t^0 = \varepsilon_2$. If $\varepsilon_1 \le (C_t^1 + C_t^2)/2 < \varepsilon_2$, we set $\boldsymbol{Th}_t^0 = (C_t^1 + C_t^2)/2$ without loss of generality. To be consistent with the statistical analysis of input data from MSH-SITS (i.e., typical examples of pixel intensity in NIRB and SAR images in Fig. 4) and imaging characterization typified by comparatively low pixel intensities of water bodies in MSH-SITS [4], [48], we set $\varepsilon_1 = (V_{\max} - V_{\min})/8$ and $\varepsilon_2 = (V_{\max} - V_{\min})/4$ for our method.

Next, we iteratively optimize flood detection results $\boldsymbol{D}_t^{\text{iter}}$ based on the minimization process of the SMRF energy function $U_t(\cdot)$. Minimizing $U_t(\cdot)$ ensures that the classification deviation between flood and nonflood areas reaches a local minimum. The SMRF energy function $U_t(\cdot)$ [25], [31] is defined as

$$U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right) = U_{\text{data}}\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right) + \theta U_{\text{context}}\left(\boldsymbol{D}_t^{\text{iter}}\right). \quad (2)$$

Note that different from the MRF energy function with the fixed receptive fields of $U_{\text{context}}(\cdot)$ in [31] [see Fig. 5(a)], our $U_{\text{context}}(\cdot)$ uses the energy-adaptive sparse SNS [see Fig. 5(b)] to expand or shrink the receptive fields of flood detection. In (2), we set $\theta = 1.6$ as a hyperparameter to balance the contributions of $U_{\text{data}}(\cdot)$ and $U_{\text{context}}(\cdot)$.

Then, the SNS in SMRF is updated. In order to make SNS in SMRF dynamically change according to the SMRF energy function, we define the energy decay ratio EDR_t to distinguish the increase and decrease of $U_t(\cdot)$

$$\text{EDR}_t = \left| \frac{U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}-1}\right) - U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right))}{U_t\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}-1}\right)} \right|. \quad (3)$$

When EDR_t decreases, $\boldsymbol{\Psi}_t$ and $\boldsymbol{D}_t^{\text{iter}}$ tend to be consistent and we should select smaller the window size of SNS to speed up the iterative convergence; otherwise, we should increase the window size of SNS to expand the receptive fields of local flood detection optimization. We initialize SNS in SMRF as $W_t^0 \times W_t^0$, where $W_t^0 = 5$. Considering the computation complexity and local optimization capability, when $\text{EDR}_t \le \varepsilon_3$, we set $W_t^{\text{iter}} = 3$ to reduce SNS in SMRF; otherwise, we set $W_t^{\text{iter}} = 5$ to expand SNS in SMRF, i.e., W_t^{iter} is updated as

$$W_t^{\text{iter}} = \begin{cases} 3, & \text{EDR}_t \le \varepsilon_3 \\ 5, & \text{otherwise} \end{cases} \quad (4)$$

where $\varepsilon_3 = 0.01$.

In (2), we design $U_{\text{data}}(\cdot)$ to measure the consistence of the detected flood map determined by $\boldsymbol{D}_t^{\text{iter}}$ with the true distributions of flood and nonflood areas in $\boldsymbol{\Psi}_t$. A lower value of $U_{\text{data}}(\cdot)$ means the better consistency between $\boldsymbol{\Psi}_t$ and $\boldsymbol{D}_t^{\text{iter}}$. $U_{\text{data}}\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right)$ is formulated as

$$U_{\text{data}}\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right) = U_{\text{data}}^{(1)}\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right) + U_{\text{data}}^{(2)}\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right) \quad (5)$$

$$U_{\text{data}}^{(1)}\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right) = \sum_{1 \le i \le M} \sum_{1 \le j \le N} \left\{ \frac{1}{2}\ln\left|2\pi\sigma_{\text{nf}}^2\right| + \frac{1}{2}(\boldsymbol{\Psi}_t(i,j) - \mu_{\text{nf}})^2\left[\sigma_{\text{nf}}^2\right]^{-1} \right\} \quad \text{if } \boldsymbol{D}_t^{\text{iter}}(i,j) = 1 \quad (6)$$

$$U_{\text{data}}^{(2)}\left(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}}\right) = \sum_{1 \le i \le M} \sum_{1 \le j \le N} \left\{ \frac{1}{2}\ln\left|2\pi\sigma_{\text{f}}^2\right| + \frac{1}{2}(\boldsymbol{\Psi}_t(i,j) - \mu_{\text{f}})^2\left[\sigma_{\text{f}}^2\right]^{-1} \right\} \quad \text{if } \boldsymbol{D}_t^{\text{iter}}(i,j) = 0 \quad (7)$$

where $\{\sigma_{\text{nf}}^2, \mu_{\text{nf}}\}$ and $\{\sigma_{\text{f}}^2, \mu_{\text{f}}\}$ are derived from $\boldsymbol{\Psi}_t$ via expectation–maximization (EM) algorithm [51] under the assumption of independence between flood and nonflood areas. Here, we start the EM iterative process for $\boldsymbol{\Psi}_t$ based on the initialized $\boldsymbol{D}_t^0$ using (1). σ_{nf}^2 and σ_{f}^2 denote the variances of pixel intensity in the flood and nonflood areas, respectively. μ_{nf} and μ_{f} represent the averaged pixel intensities of flood and nonflood areas, respectively. The Gibbs energy function $U_{\text{context}}(\cdot)$ [31] in (2) is used as the local consistency measure for the flood detection results in the SNS of SMRF. When the local similarity of pixels becomes better, the value of $U_{\text{context}}(\cdot)$ is smaller. Otherwise, the value of $U_{\text{context}}(\cdot)$ is bigger. Specifically, $U_{\text{context}}\left(\boldsymbol{D}_t^{\text{iter}}\right)$ is defined as

$$U_{\text{context}}\left(\boldsymbol{D}_t^{\text{iter}}\right) = \sum_{1 \le i \le M} \sum_{1 \le j \le N} \left\{ \sum_{(g,h) \in \beta(i,j)} \delta_k\left(\boldsymbol{D}_t^{\text{iter}}(i,j), \boldsymbol{D}_t^{\text{iter}}(g,h)\right) \right\} \quad (8)$$

where $\beta(i,j)$ represents the set of neighborhood pixels (see blue pixels in Fig. 5) around the current pixel (see the red pixel in Fig. 5), $(g,h) \ne (i,j)$, and the Kronecker delta function δ_k [34] is defined as

$$\delta_k\left(\boldsymbol{D}_t^{\text{iter}}(i,j), \boldsymbol{D}_t^{\text{iter}}(g,h)\right) = \begin{cases} -1, & \text{if } \boldsymbol{D}_t^{\text{iter}}(i,j) = \boldsymbol{D}_t^{\text{iter}}(g,h) \\ 0, & \text{if } \boldsymbol{D}_t^{\text{iter}}(i,j) \ne \boldsymbol{D}_t^{\text{iter}}(g,h) \end{cases} \quad (9)$$

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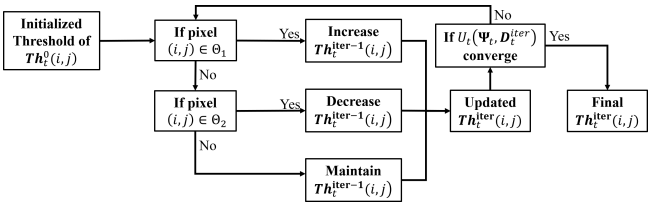


Fig. 6. Flowchart of the updating process of $\boldsymbol{Th}_t^{\text{iter}}(i, j)$.

In the iterative process, we aim to seek the local minimum of $U_t(\cdot)$, which is accomplished by updating the dynamic thresholds $\boldsymbol{Th}_t^{\text{iter}}$ [25], [52], [53]. We take the contextual features of a pixel into account to achieve local detection optimization because a pixel is more likely to be affected by its surrounding pixels. Specifically, we extract the edge contour $\boldsymbol{Bd}_t^{\text{iter}-1} \in \mathcal{R}^{M\times N}$ of $\boldsymbol{D}_t^{\text{iter}-1}$, where $\boldsymbol{Bd}_t^{\text{iter}-1}(i, j) = 1$ denotes the edge contour points of flood areas and $\boldsymbol{Bd}_t^{\text{iter}-1}(i, j) = 0$ represents the noncontour pixels. We only update these pixels around the edge contour to diminish the value of $U_{\text{data}}(\cdot)$ which speed up the convergence of $U_t(\cdot)$. We employ the global Manhattan distance (MD) matrix $\boldsymbol{d}_t^{\text{iter}-1}$ as the spatial constraints of $\boldsymbol{Th}_t^{\text{iter}}$

$$\boldsymbol{d}_t^{\text{iter}-1}(i, j) = M((i, j), (i_k, j_k)), \quad \text{if } \boldsymbol{Bd}_t^{\text{iter}-1}(i, j) = 0 \tag{10}$$

where (i_k, j_k) represents the coordinates of "1" elements in $\boldsymbol{Bd}_t^{\text{iter}-1}$, $M(\cdot)$ represents the minimum of the MDs between the pixel with $\boldsymbol{Bd}_t^{\text{iter}-1}(i, j) = 0$ and all the pixels with label 1 in $\boldsymbol{Bd}_t^{\text{iter}-1}$. A large value of $\boldsymbol{d}_t^{\text{iter}-1}(i, j)$ means that the pixel (i, j) in $\boldsymbol{\Psi}_t$ is far away from the contour of flood areas. In [54], the river continuum concept (RCC) is proposed to describe that the flood area is closely connected. Therefore, the detected isolated and small flood areas that are close to flood contours should be expanded and merged into the main flood areas. We choose to update $\boldsymbol{Th}_t^{\text{iter}}(i, j)$ based on the voting results of SNS to enlarge the detected flood areas or shrink the suspected flood areas, which iteratively reduces the value of $U_t(\cdot)$ to a local minimum. Specifically, the detailed updating flowchart of $\boldsymbol{Th}_t^{\text{iter}}$ is given in Fig. 6 and $\boldsymbol{Th}_t^{\text{iter}}$ is updated as

$$\boldsymbol{Th}_t^{\text{iter}}(i, j) = \begin{cases} \left(\boldsymbol{Th}_t^{\text{iter}-1}(i, j) + 1\right) \times 1.01, & \text{if } (i, j) \in \Theta_1 \\ \left(\boldsymbol{Th}_t^{\text{iter}-1}(i, j) + 1\right)/1.1, & \text{if } (i, j) \in \Theta_2 \\ \boldsymbol{Th}_t^{\text{iter}-1}(i, j), & \text{otherwise} \end{cases} \tag{11}$$

where $\Theta_1 = \{(i, j) | \text{if } \boldsymbol{D}_t^{\text{iter}-1}(i, j) = 1, \boldsymbol{d}_t^{\text{iter}-1}(i, j) < \varepsilon_4, \boldsymbol{\Psi}_t(i, j) < \left(1.1 \times \boldsymbol{Th}_t^0(i, j)\right)$, and $\beta_{\text{f}} \geq (0.3 \times \left(W_t^{\text{iter}}\right)^2)\}$, $\Theta_2 = \{(i, j) | \text{ if } \boldsymbol{d}_t^{\text{iter}-1}(i, j) < \varepsilon_4, \boldsymbol{\Psi}_t(i, j) > \left(1.1 \times \boldsymbol{Th}_t^0(i, j)\right)$, and $\beta_{\text{nf}} > (0.7 \times \left(W_t^{\text{iter}}\right)^2)\}$, and β_{f} and β_{nf} denote the number of flood and nonflood pixels in SNS, respectively. In (11), the parts of "×1.01" and "/1.1" are used to increase the flood threshold of each pixel and suppress the flood threshold of pixels that are most likely to be nonflood areas, respectively. According to RCC [54], the pixel set Θ_1 represents a small range of floods around the reliable flood detection results and should be raised. Θ_2 denotes the set of pixels far away from the reliable flood areas that ought to be reduced or removed. We set $\varepsilon_4 = 8$ in (11) based on the comprehensive consideration of spatial constraints and computational complexity. Then, we update $\boldsymbol{D}_t^{\text{iter}}$ based on $\boldsymbol{Th}_t^{\text{iter}}$.

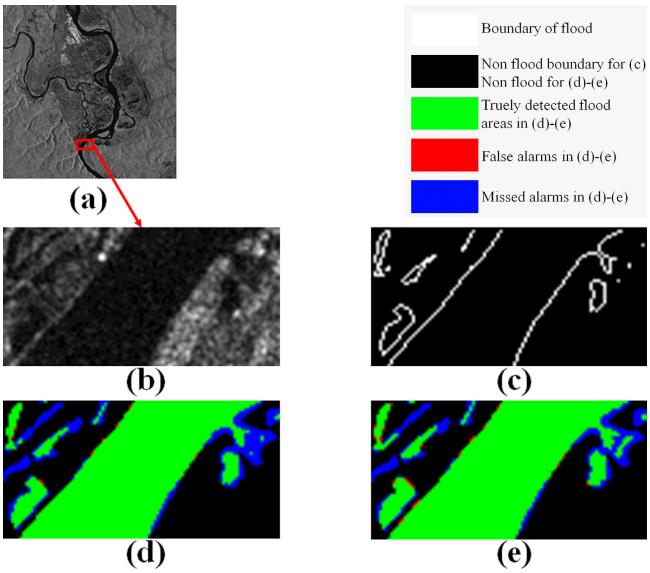


Fig. 7. Visualization of the updating strategy of $\boldsymbol{Th}_t^{\text{iter}}$: (a)–(e) thumbnails of $\boldsymbol{\Psi}_1$, the small region of $\boldsymbol{\Psi}_1$ [$\boldsymbol{\Psi}_1(R)$], the boundary of $\boldsymbol{\Psi}_1(R)$ [$\boldsymbol{Bd}_1^0(R)$], the flood map of $\boldsymbol{D}_1^0(R)$, and the final spatial clustering results of $\boldsymbol{\Psi}_1(R)[\boldsymbol{D}_1(R)]$, respectively, where "$(R)$" denotes the small region of the target scene. White: boundary of the detected flood areas in (c); black: background in (c)/correct detection of nonflood areas in (d) and (e); green: truly detected flood areas; red: false detection of nonflood areas; and blue: miss detection of flood areas.

To describe (10) and (11) more intuitively, we show the detailed iterative flood detection process based on dynamic updating of $\boldsymbol{Th}_t^{\text{iter}}$ in the partial of $\boldsymbol{\Psi}_1$ (see Fig. 7), where we only update the thresholds of pixels around the edge contours of the detected flood regions [see Fig. 7(c)]. In the iterative process of (10) and (11), we choose to increase/decrease/maintain the flood threshold of each pixel based on the subordination between pixels in SNS of SMRF and pixels in $\{\Theta_1, \Theta_2\}$. As shown in Fig. 7(d) and (e), we obtain the improved flood detection results without a noticeable increase of false alarms (i.e., red areas) using (10) and (11), where the correct detected flood areas (i.e., green areas) are expanded.

In each iteration of spatial clustering, we update the formulas of (2)–(11). The termination condition of the iterative process is defined as $\{\text{if}(U_t(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}-1}) - U_t(\boldsymbol{\Psi}_t, \boldsymbol{D}_t^{\text{iter}})) < 0$ or iter = itermax$\}$. When the iterations stop, the local optimum of $U_t(\cdot)$ is reached. In order to intuitively represent the flood area, $\boldsymbol{D}_t = \mathbf{1} - \boldsymbol{D}_t^{\text{iter}-1}$ is determined as the final spatial clustering results, where all the elements in $\mathbf{1} \in \mathcal{R}^{M\times N}$ is 1. Here, we set itermax = 100 without loss of generality. In Fig. 8, we show the convergence of the SMRF energy values in the iterative process.

D. Temporal Fusion

The temporal fusion step is summarized in Algorithm 2, which is designed to attenuate the effects of false alarms encountered in the single remote sensing image. We exploit

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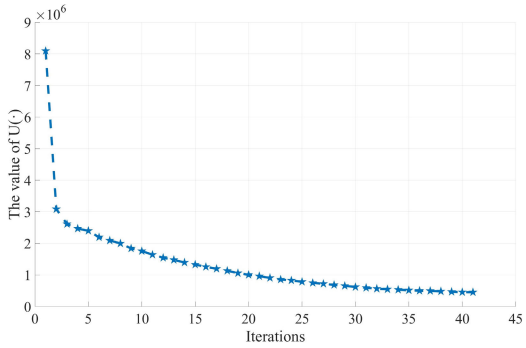


Fig. 8. Convergence of the SMRF energy value of (2) in the iterative process, taking $\boldsymbol{\Psi}_9$ as an example.

Algorithm 2 Temporal Fusion Step

Input: The spatial clustering results $\boldsymbol{D} = \{\boldsymbol{D}_t\}_{1\le t\le s}$ derived from the Algorithm 1.

for $t = 1 : S$ **do**

  Initialization: Let $\boldsymbol{iter}_t$ be the historical information initialized using (12). let $Thresh_t$ reprsent the global flood area proportion of $\boldsymbol{\Psi}_t$ and $\boldsymbol{D}_t(k) \in \mathcal{R}^{M\times N}$ represent the k th four-connected regions in $\boldsymbol{D}_t$ (only the flood pixels in the k th connected region are marked as "1" and the rest pixels are marked as "0" in $\boldsymbol{D}_t(k)$), for $k = 1, 2, \ldots, num_t$, where num_t indicates the totality of 4-connected areas in $\boldsymbol{D}_t$.

  for $k = 1$: num_t do

  Calculate the overlap ratio $\alpha_k = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}(\boldsymbol{D}_t(k)\odot \boldsymbol{Incre}_t)(i,j)}{\sum_{i=1}^{M}\sum_{j=1}^{N}(\boldsymbol{D}_t(k))(i,j)}$ between $\boldsymbol{D}_t(k)$ and $\boldsymbol{Incre}_t$

  If $\alpha_k > Thresh\ h_t$, the "1" elements in $\boldsymbol{D}_t(k)$ are determined as the reliable flood area in $\boldsymbol{STF}_t(k) \in \mathcal{R}^{M\times N}$. All the $\boldsymbol{STF}_t(k), \forall k$ formulate the flood detection results $\boldsymbol{STF}_t \in \mathcal{R}^{M\times N}$ of $\boldsymbol{\Psi}_t$, [see (13)-(14)]

  end

end

Output: $\boldsymbol{STF} = \{\boldsymbol{STF}_t\}_{(1\le t\le S)}$ and the STFCD results $\boldsymbol{CD}_{t\to(t-l)}$ between $\bar{\boldsymbol{\Psi}}_t$ and $\boldsymbol{\Psi}_{t-l}$ for $1 \le l < t$, [sec(15)]

the complementary characteristics in MSH-SITS, i.e., the flood detection results from SAR images can help to weaken the interferences of cloud shadows and terrain shadows in flood detection results based on the optical images and the flood detection results derived from the optical images can help to decrease the conflicts of terrain shadows in the flood detection results of the SAR images. We employ the historical information $\boldsymbol{Incre}_t \in \mathcal{R}^{M\times N}$ of the spatial clustering results as constraints to realize the cross-reduction of the effect from terrain shadows in SAR images and cloud shadows and topography shadows in optical images for flood detection. First, we initialize $\boldsymbol{Incre}_t$ and the global flood area proportion of $\boldsymbol{\Psi}_t$ as the temporal fusion constraints. Then, under the constraints of $\boldsymbol{Incre}_t$, the flood detection results $\boldsymbol{STF}_t$ are obtained by reducing the suspected flood areas in $\boldsymbol{D}_t$. Finally, the CD results using the STFCD algorithm are generated by comparing the spatiotemporally fused flood detection results $\boldsymbol{STF}_t$ of the specific two images.

First, if and only if the accumulated number of MSH-SITS images exceeds 3, the historical information $\boldsymbol{Incre}_t$ is initialized as temporal constraints

$$\boldsymbol{Incre}_t(i,j) = \begin{cases} 1, & \text{if } \boldsymbol{A}_t(i,j) \ge 2 \text{ and } \omega \le 6 \text{ days} \\ 1, & \text{if } \boldsymbol{B}_t(i,j) \ge 1 \text{ and } \omega > 6 \text{ days} \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\boldsymbol{A}_t = \boldsymbol{D}_t \odot (\boldsymbol{D}_{t-1} + \boldsymbol{D}_{t-2} + \boldsymbol{D}_{t-3})$ and $\boldsymbol{B}_t = \boldsymbol{D}_t \odot (\boldsymbol{D}_{t-1} + \boldsymbol{D}_{t-2})$. In (12), considering the rapid change of flood areas, we set ω as the acquisition interval constraints between $\boldsymbol{\Psi}_t$ and $\boldsymbol{\Psi}_{t-3}$ to dynamically adjust $\boldsymbol{Incre}_t$. When $\omega > 6$ days, the spatial clustering results of the two images from MSH-SITS obtained before the current time t are employed as constraints. When $\omega \le 6$ days, the spatial clustering results of the three images from MSH-SITS acquired before the current time are used as constraints. In (12), a large element in $\boldsymbol{A}_t$ or $\boldsymbol{B}_t$ means that the current pixel location is occupied by flood areas.

Second, we reduce the suspected flood area using the constraint of $\boldsymbol{Incre}_t$. Changes in the flood zone often occur continuously over a densely continuous observation of the same area under RCC [54]. Specifically, we compare the four-connected flood regions (regions consist of four-connected flood pixels) between $\boldsymbol{D}_t$ and $\boldsymbol{Incre}_t$ [55]. Specifically, we define $\boldsymbol{D}_t(k) \in \mathcal{R}^{M\times N}$, the k th four-connected matrix in $\boldsymbol{D}_t$, where "1" labeled pixels in $\boldsymbol{D}_t(k)$ represent the kth flood regions in $\boldsymbol{D}_t$, "0" labeled pixels in $\boldsymbol{D}_t(k)$ are the other areas in $\boldsymbol{D}_t$, $k = 1, \ldots,$ num$_t$, and num$_t$ denotes the total number of the four-connected flood regions in $\boldsymbol{D}_t$. Next, $\boldsymbol{D}_t(k)$ is determined as the reliable flood area of $\boldsymbol{\Psi}_t$ if the overlap ratio $\alpha_k = (\sum_{i=1}^{M}\sum_{j=1}^{N}(\boldsymbol{D}_t(k) \odot \boldsymbol{Incre}_t)(i,j) / \sum_{i=1}^{M}\sum_{j=1}^{N}(\boldsymbol{D}_t(k))(i,j))$ between $\boldsymbol{D}_t(k)$ and $\boldsymbol{Incre}_t$ is larger than Thresh$_t$ $= (\sum_{i=1}^{M}\sum_{j=1}^{N}\boldsymbol{D}_t(i,j)/M \times N)$ (the global flood area proportion of $\boldsymbol{\Psi}_t$). The final flood detection results of $\boldsymbol{\Psi}_t$ with STF are formulated as

$$\boldsymbol{STF}_t = \sum_{1\le k\le \text{num}_t} \boldsymbol{STF}_t(k) \tag{13}$$

$$\boldsymbol{STF}_t(k) = \begin{cases} \boldsymbol{D}_t(k), & \alpha_k > \text{Thresh}_t \\ \boldsymbol{0}, & \text{otherwise.} \end{cases} \tag{14}$$

As shown in Fig. 9, the false flood detection rate is significantly reduced by employing the historical information $\boldsymbol{Incre}_t$.

Finally, the CD results of $\boldsymbol{\Psi} = \{\boldsymbol{\Psi}_t\}_{1\le t\le S}$ are obtained. Aiming to acquire the change map $\boldsymbol{CD}_{t\to(t-l)}$ between $\boldsymbol{\Psi}_t$ and $\boldsymbol{\Psi}_{t-l}$, where $1 \le l < t$. When $(\boldsymbol{STF}_t - \boldsymbol{STF}_{t-l})(i,j) > 0$, we determine the pixel $\boldsymbol{CD}_{t\to(t-l)}(i,j) = 1$ as changed flood pixel; otherwise, we set $\boldsymbol{CD}_{t\to(t-l)}(i,j) = 0$ as nonchanged pixel, i.e.,

$$\boldsymbol{CD}_{t\to(t-l)}(i,j) = \begin{cases} 1, & (\boldsymbol{STF}_t - \boldsymbol{STF}_{t-l})(i,j) > 0 \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

E. Experimental Setup and Evaluation Indicators

We examined the effectiveness of the proposed algorithm on 12 different images of MSH-SITS, including seven SAR images and five optical images in HT, China, 2021

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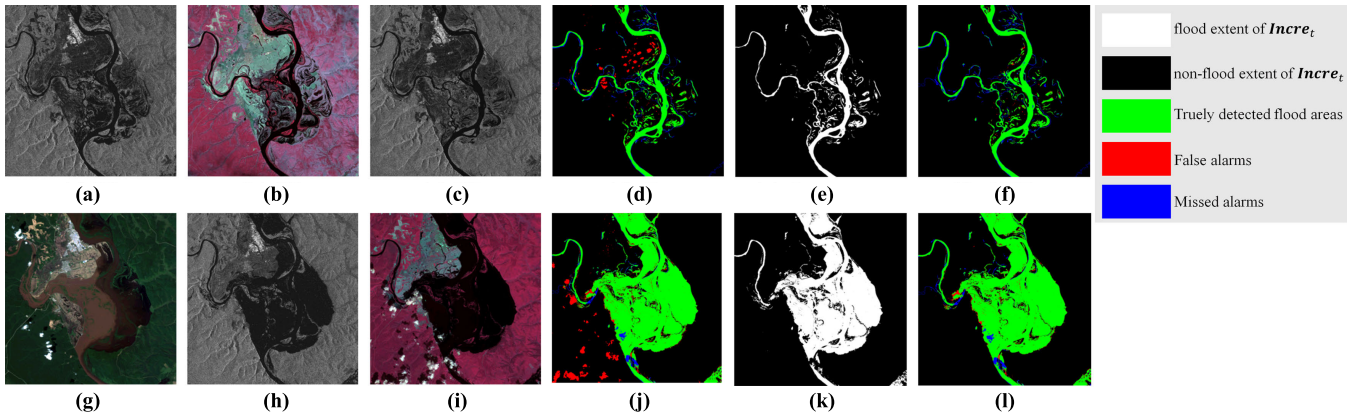


Fig. 9. Visual examples of the temporal fusion step by our STF-based strategy: (a)–(f) Ψ_2, Ψ_3, Ψ_4, spatial clustering results (D_4) of Ψ_4, historical information ($Incre_4$) originated from Ψ_2 and Ψ_3, and flood detection results (STF_4) of Ψ_4, respectively, and (g)–(l) Ψ_9, Ψ_{10}, Ψ_{11}, D_{11}, $Incre_{11}$, and the final flood detection results (STF_4) of Ψ_{11}, respectively. In the comparison of D_t and STF_t, the false flood detection rate is significantly reduced by employing the historical information $Incre_t$.

(see Table II). First of all, we prove the effectiveness of the continuing global and partial flood detection capabilities of our STF-based strategy in Sections IV-A and IV-B, respectively, which is a prerequisite for obtaining reliable continuous CD results. Then, our STFCD with SMRF is compared with seven state-of-the-art CD algorithms on the flood extents CD task in Section IV-C. These seven CD algorithms are implemented by the default parameters in the deep homogeneous feature fusion (DHFF) algorithm [56], HPT algorithm [16], IFBCD algorithm [25], principal component analysis network (PCANet) [57], semisupervised Siamese network (S^3N) [17], symmetric convolutional coupling network (SCCN) [19], and our STFCD with MRF [25], respectively. Since NIRBs and SAR images have similar low pixel intensity characteristics in flood areas [4], [48], we employ the same strategy to delineate flood areas in optical and SAR images of MSH-SITS in accordance with Fig. 2(a), ASTER Global DEM V3 [46], and MSH-SITS. Specifically, we use ENVI software to discard the blurry areas between high and low scattering/reflection intensity on the mixed pixels along the flood boundary and set the specific area with sediments suspended or emergent vegetation as flood areas when the elevation of the specific area is less than or equal to that of the surrounding flood areas. All the preprocessing steps, our algorithms, and the other comparative experiments included in this work were performed on the same platform, which is implemented with 2 Intel[1] Xeon Platinum 8280 CPUs at 2.7 GHz (two CPUs, 56 cores), 1 TB of RAM, and two NVIDIA TITAN RTX graphic processing units (GPUs) of 48 GB.

Inasmuch as many studies have included the overall classification accuracy (OA) [58] in the evaluation indicators, we also use OA as one of our evaluation indicators. However, OA is not reliable in scenarios with sparse change pixels. We further select two comprehensive evaluation indexes, i.e., the Kappa coefficient (KC) [59] and the first error measure F_1 [60], as our evaluation indicators, which can directly reflect the flood detection performance with higher sensitivity to the false alarms and missed alarms in sparse scenes than OA. The larger value of OA, KC, and F_1 mean the higher classification accuracy of flood and nonflood areas.

[1]Trademarked.

IV. Results and Discussion

A. Analysis of Global Flood Maps in HT

We define all the flood areas contained in the image slices corresponding to the main study area in MSH-SITS (see Fig. 1) as global flood maps. In this section, we evaluate global flood maps derived from MSH-SITS using our STF-based strategy, i.e., the combination of the spatial clustering step (see Section III-C) and the temporal fusion step (see Section III-D) without the CD analysis, considering the influences of cloud shadows and topography shadows in MSH-SITS. We first compare the statistic indicators between our STF-based strategy and the other seven competitive algorithms (i.e., eight unsupervised algorithms) on $\Psi_1 - \Psi_{12}$ in HT (see Table III). Then, we add two supervised neural network-based segmentation algorithms, i.e., Unet++ [61] and XBound-Former (XBF) [62], as competitive methods and conduct numerical and visual analysis based on ten methods on the global flood maps of $\Psi_{11} - \Psi_{12}$, respectively (see Table IV and Fig. 10).

In general, we focus on the delineation of the outbreak of large-scale floods that can easily evolve into flood disasters caused by river expansion. Since the generation of the global flood maps is also of crucial importance for flood assessment, we challenge the problem of global flood maps generation on MSH-SITS datasets in HT. In Table III, we compare the global flood detection performance of eight unsupervised algorithms, i.e., the FCM algorithm [63], simple linear iterative clustering (SLIC) algorithm [64], MRF [25], our proposed spatial clustering step (S-step), FCM combined with temporal fusion step (FCM-TF), SLIC integrated with temporal fusion step (SLIC-TF), MRF linked with temporal fusion step (MRF-TF), and our proposed STF on the dataset. As shown in Table III, using FCM [63], SLIC [64], and MRF [25] to directly perform binary classification of flood maps is not stable, due to the interference from terrain shadows and lighting conditions, in long-term flood detection, while our S-step algorithm

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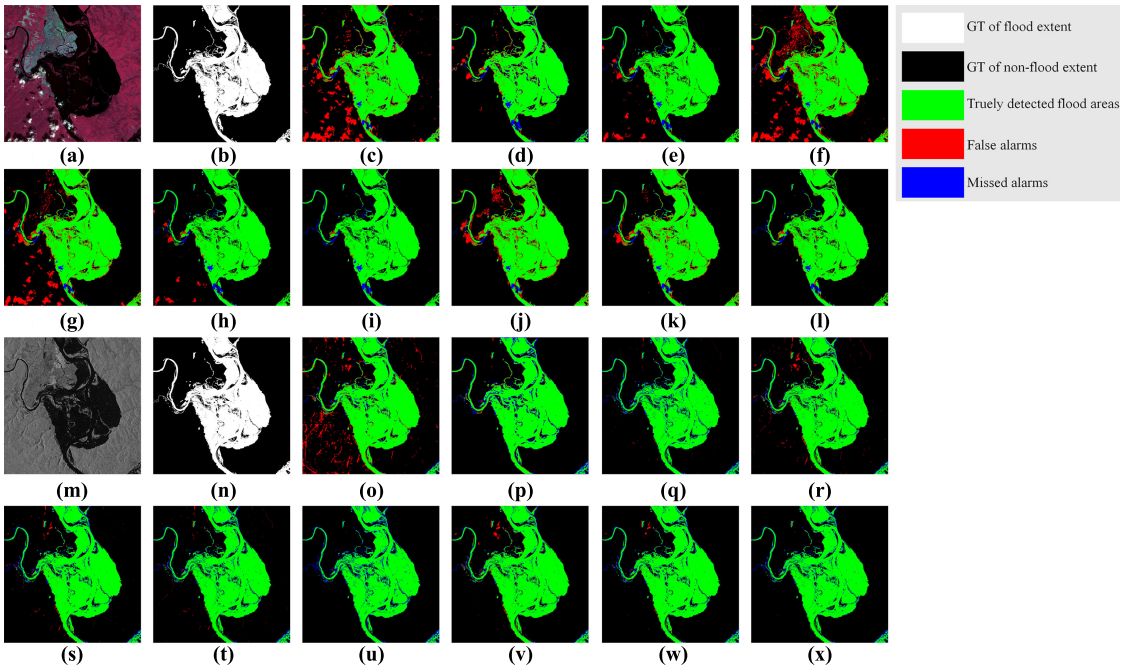


Fig. 10. Comparison of flood maps corresponding to Ψ_{11} and Ψ_{12} based on ten algorithms: (a)–(l) Ψ_{11}, GT of the global flood map of Ψ_{11} ($\boldsymbol{GT}_{11}$), the global flood map conducted by Unet++ ($\boldsymbol{Unet++}_{11}$), XBF ($\boldsymbol{XBF}_{11}$), FCM ($\boldsymbol{FCM}_{11}$), SLIC ($\boldsymbol{SLIC}_{11}$), MRF ($\boldsymbol{MRF}_{11}$), S-step ($\boldsymbol{S\text{-}step}_{11}$), FCM-TF ($\boldsymbol{FCM\text{-}TF}_{11}$), SLIC-TF ($\boldsymbol{SLIC\text{-}TF}_{11}$), MRF-TF ($\boldsymbol{MRF\text{-}TF}_{11}$), and our STF ($\boldsymbol{STF}_{11}$), respectively, and (m)–(x) Ψ_{12}, $\boldsymbol{GT}_{12}$, $\boldsymbol{Unet++}_{12}$, $\boldsymbol{XBF}_{12}$, $\boldsymbol{FCM}_{12}$, $\boldsymbol{SLIC}_{12}$, $\boldsymbol{MRF}_{12}$, $\boldsymbol{S\text{-}step}_{12}$, $\boldsymbol{FCM\text{-}TF}_{12}$, $\boldsymbol{SLIC\text{-}TF}_{11}$, $\boldsymbol{MRF\text{-}TF}_{11}$, and our $\boldsymbol{STF}_{12}$, respectively.

TABLE III
EVALUATION OF FLOOD MAPS OF Ψ_1-Ψ_{12} USING EIGHT ALGORITHMS

Methods \ Indicators	Flood maps of Ψ_1-Ψ_{12}		
	OA	KC	F_1
FCM [63]	0.9090 ± 0.0683	0.7019 ± 0.2194	0.7482 ± 0.1927
SLIC [64]	0.8795 ± 0.0728	0.6542 ± 0.2133	0.7185 ± 0.1863
MRF [25]	0.9304 ± 0.0499	0.7712 ± 0.1598	0.8123 ± 0.1373
our S-step	0.9766 ± 0.0074	0.9053 ± 0.0476	0.9195 ± 0.0458
FCM-TF	0.9179 ± 0.0684	0.7242 ± 0.2243	0.7656 ± 0.1965
SLIC-TF	0.8897 ± 0.0721	0.6755 ± 0.2149	0.7343 ± 0.1871
MRF-TF	0.9417 ± 0.0426	0.7985 ± 0.1407	0.8334 ± 0.1210
Our STF	**0.9784 ± 0.0047**	**0.9111 ± 0.0345**	**0.9241 ± 0.0344**

TABLE IV
EVALUATION OF FLOOD MAPS OF Ψ_{11}, $-\Psi_{12}$ BASED ON TEN ALGORITHMS

Methods \ Indicators	Flood maps of Ψ_{11} and Ψ_{12}		
	OA	KC	F_1
FCM [63]	0.9723	0.9314	0.9506
SLIC [64]	0.9483	0.8782	0.9150
MRF [25]	0.9596	0.9031	0.9316
our S-step	0.9769	0.9427	0.9587
Unet++ [61]	0.9403	0.8589	0.9014
XBF [62]	0.9631	0.9088	0.9345
FCM-TF	0.9776	0.9440	0.9595
SLIC-TF	0.9664	0.9186	0.9423
MRF-TF	0.9725	0.9327	0.9518
Our STF	**0.9790**	**0.9478**	**0.9624**

maintains stable and superior flood detection performance benefiting from our proposed SMRF algorithm. Encouragingly, with the TF strategy, the flood detection performances of FCM-TF, SLIC-TF, MRF-TF, and STF are all significantly improved than FCM [63], SLIC [64], MRF [25], and our S-step, respectively, which proofs the effectiveness of our TF-based strategy. Moreover, our STF-based strategy obtains relatively better flood detection performance on $\Psi_1 - \Psi_{12}$ than the other seven methods in Table III.

In the following experiments, we utilize augmented $\Psi_1 - \Psi_{10}$ as the training set and the validation set (1440 slices with the ratio of 0.9393:0.0707) for both Unet++ [61] and XBF [62] and apply $\Psi_{11} - \Psi_{12}$ as the testing images for all the ten algorithms. In addition, Unet++ and XBF algorithms only apply $\Psi_1 - \Psi_{10}$ as training samples. Unet++ and XBF algorithms leave the temporal correlation of MSH-SITS out of consideration, while FCM-TF, SLIC-TF, MRF-TF, and our STF take the flood detection results based on historical satellite images (recent two–three satellite images) as the constraints for flood detection based on the current satellite image. Specifically, we employ $\boldsymbol{Incre}_{11}$ derived from $\boldsymbol{D}_9$ and $\boldsymbol{D}_{10}$ and $\boldsymbol{Incre}_{12}$ originated from $\boldsymbol{D}_9 - \boldsymbol{D}_{11}$ as historical information for our STF-based strategy when testing Ψ_{11} and Ψ_{12}, respectively.

The experimental results on Ψ_{11} and Ψ_{12} are shown in Fig. 10 and Table IV. The black, white, green, red, and blue colors in Fig. 10 represent nonflood area, flood area, true positive, false alarms, and missed alarms, respectively. As shown in Fig. 10(c)–(l), $\boldsymbol{Unet++}_{11}$, $\boldsymbol{FCM}_{11}$, $\boldsymbol{SLIC}_{11}$, $\boldsymbol{MRF}_{11}$,

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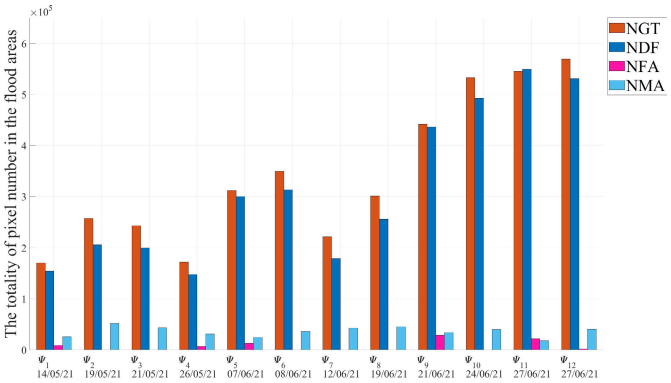


Fig. 11. Evaluation statistics of flood maps obtained by our STF-based strategy in $\boldsymbol{\Psi}_1 - \boldsymbol{\Psi}_{12}$. The yellowish-brown bar denotes the GT of the number of flood pixels in the specific image (NGT). The blue bar (NDF), the pink bar (NFA), and the sky-blue bar (NMA) represent the quantity of overall detected flood pixels, detected flood pixels belonging to false alarms, and missed flood pixels, respectively.

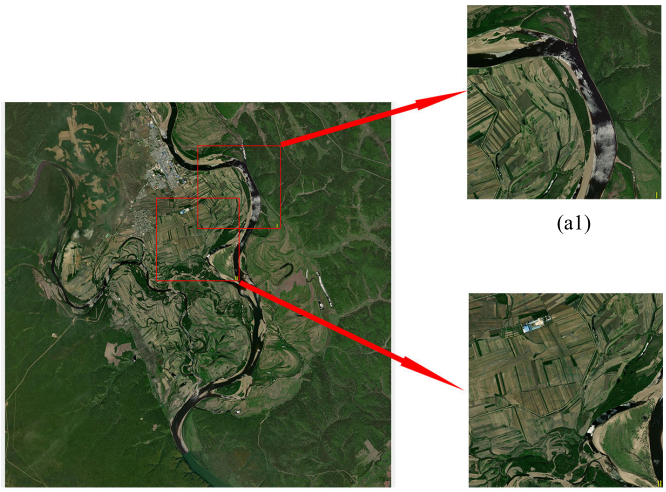


Fig. 12. Overview of R1 in HT and R2 in HT: (a) ORGB; (a1) R1 in ORGB; and (a2) R2 in ORGB.

and $\boldsymbol{S-step}_{11}$ are severely disturbed by cloud shadows in optical images, while $\boldsymbol{XBF}_{11}$, $\boldsymbol{FCM-TF}_{11}$, $\boldsymbol{SLIC-TF}_{11}$, $\boldsymbol{MRF-TF}_{11}$, and $\boldsymbol{STF}_{11}$ have significantly lighten the noise effects from the cloud shadows. From Fig. 10(o)–(x), we can observe that $\boldsymbol{Unet++}_{12}$, $\boldsymbol{XBF}_{12}$, $\boldsymbol{FCM}_{12}$, $\boldsymbol{SLIC}_{12}$, $\boldsymbol{MRF}_{12}$, and $\boldsymbol{S-step}_{12}$ introduce more false alarms than $\boldsymbol{SLIC-TF}_{12}$, $\boldsymbol{FCM-TF}_{12}$, $\boldsymbol{MRF-TF}_{12}$, and $\boldsymbol{STF}_{12}$, while $\boldsymbol{STF}_{12}$ achieves relatively better flood map than the other nine methods with less missed alarms and false alarms. In Table IV, our TF-based methods (i.e., FCM-TF, SLIC-TF, MRF-TF, and STF algorithms) achieve higher OA, KC, and F_1 of the flood map generation in comparison with Unet++ [61] and XBF [62] algorithms. This is because we utilize the complementary characteristics of MSH-SITS to reduce the interference of cloud shadows in optical images and terrain shadows in SAR images on flood detection. Particularly, our proposed STF algorithm obtains better flood detection performance than the other nine comparison methods (in Table IV) benefiting from our proposed STF strategy. In summary, our STF-based strategy reduces the influence of noise effects, including false alarms and missed alarms, which is accomplished by the exploitation of the spatiotemporal relevance in MSH-SITS data.

We further analyze the robustness of flood detection via our STF-based strategy in Fig. 11, where the yellowish-brown bars represent the total numbers of flood pixels in GTs (NGT), the blue bars denote the overall accurately detected flood pixels (NDF), the pink bars signify the totality of the false alarm pixels (NFA), and the sky-blue bars denote the number of pixels representing missed alarm (NMA), respectively. Fig. 11 reveals that there is steady consistency between the GTs and the flood detection results by STF-based strategy.

Based on the global flood detection results in Fig. 11, it can be found that there has been a continuous rising of the flood areas on HT in late June 2021 and the flood in HT peaked on June 27, 2021, which is consistent with the flood peak time measured by Huma HS and Sandaoqia HS in Fig. 1. According to HPHWRC, the water level of the Huma Bridge station of the tributary Huma River exceeded the warning level of 1.98 m, on 08:00, June 21, 2021 [65]. As of 14:00 (June 22, 2021), the water level of the Huma Bridge station given by HPHWRC was increasing to 2.30 m above the warning water level [66]. These flood detection results of $\boldsymbol{\Psi}_8 - \boldsymbol{\Psi}_{12}$ via STF-based strategy in Fig. 11 are remarkably consistent with the actual flood occurrence.

B. Analysis of Partial Flood Maps via the STF-Based Strategy

In this section, we specifically analyze the flood extent delineation performance of our proposed algorithm in two areas, where Region 1 (R1) is located on the southeast side of HT and adjacent to the mainstream of the Amur River and Region 2 (R2) is located in the northwest of the intersection of the Huma River and the Amur River. The plots of R1 and R2 can be found in Fig. 12. All the MSH-SITS in R1 and R2 have the same pixel size of 301 × 301 and the same spatial resolution of 16 × 16 m. Since ORGB is an optical image acquired in the dry season, we can compare and analyze the actual flood areas more intuitively based on R_1 and R_2 in the ORGB image.

The comparison between detected flood maps of $\boldsymbol{STF}_1(\boldsymbol{R}_1) - \boldsymbol{STF}_{12}(\boldsymbol{R}_1)$ and $\boldsymbol{STF}_1(\boldsymbol{R}_2) - \boldsymbol{STF}_{12}(\boldsymbol{R}_2)$ using the STF-based strategy and GTs based on of $\boldsymbol{\Psi}_1(\boldsymbol{R}_1)\boldsymbol{\Psi}_{12}(\boldsymbol{R}_1)$ and $\boldsymbol{\Psi}_1(\boldsymbol{R}_2) - \boldsymbol{\Psi}_{12}(\boldsymbol{R}_2)$ is shown in Fig. 13. Visual inspection of Fig. 13(i2)–(l2) indicates that the surge of flood areas on late June 2021 were appear adjacent to the confluence of the Huma River and the Amur River. Obviously, our proposed STF-based strategy can effectively extract large-scale flood extent while introducing small-scale false alarms since precipitation reduces the SAR backscatter intensities in the low-lying areas. Consequently, there are high false alarm rates in $\boldsymbol{STF}_1(\boldsymbol{R}_1)$, $\boldsymbol{STF}_4(\boldsymbol{R}_1)$, $\boldsymbol{STF}_1(\boldsymbol{R}_2)$, and $\boldsymbol{STF}_4(\boldsymbol{R}_2)$ in accordance with the visual inspection of Fig. 13(a1), (d1), (a2), and (d2), respectively. According to the actual weather records from the China Meteorological Data Service Center (CMDC), Beijing, China, the light rain weather in Huma County occurred on May 14 and 26, 2021

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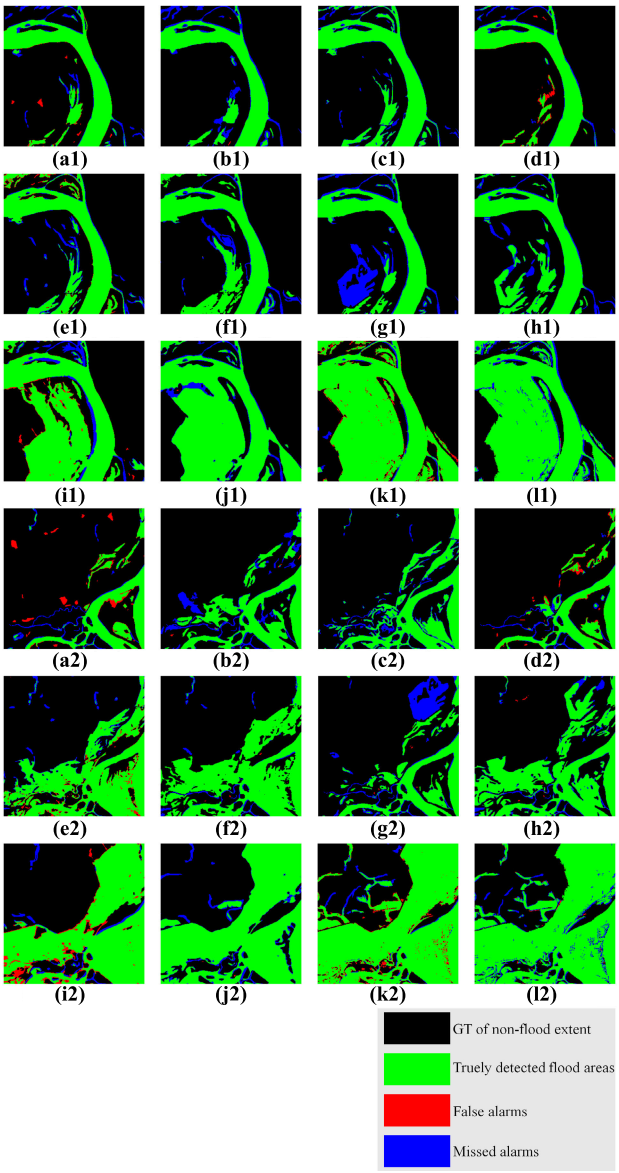


Fig. 13. Flood detection results: (a1)–(l1) $STF_1(R_1) - STF_{12}(R_1)$ and (a2)–(l2) $STF_1(R_2) - STF_{12}(R_2)$ using the proposed STF-based strategy based on $\Psi_1 - \Psi_{12}$.

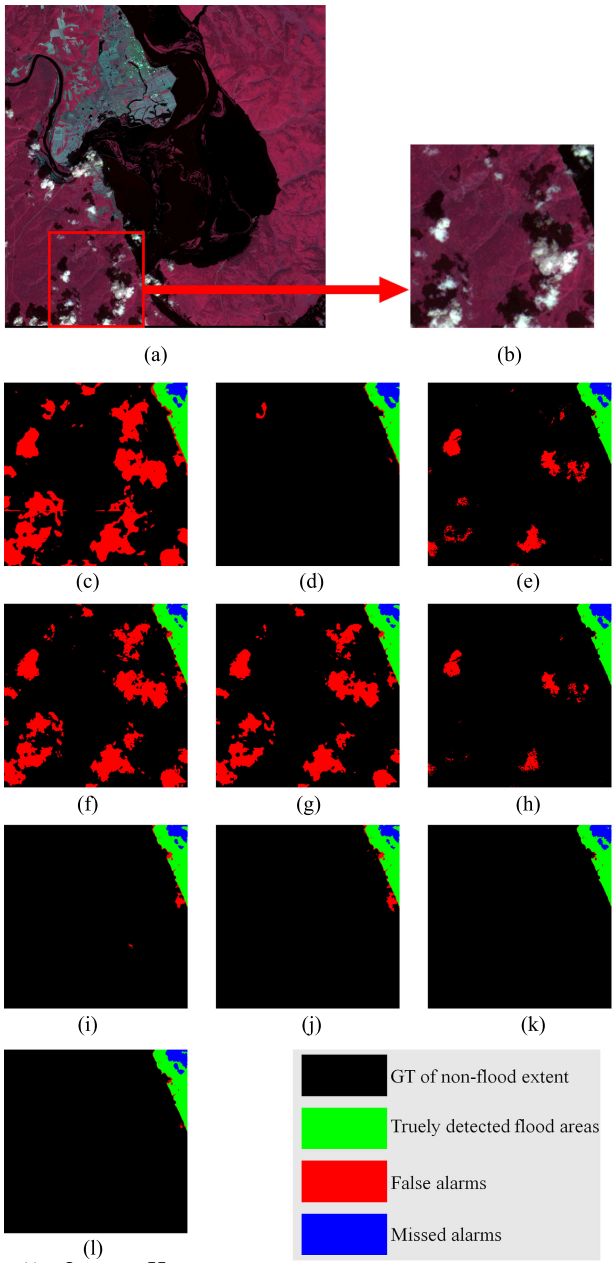


Fig. 14. Close-up analysis of false alarms caused by cloud shadows: (a) Ψ_{11}; (b) zoomed up region of cloud shadows in Ψ_{11} (R_3); (c) R_3 of $Unet++_{11}$; (d) R_3 of XBF_{11}; (e)R_3 of FCM_{11}; (f) R_3 of $SLIC_{11}$; (g) R_3 of MRF_{11}; (h) R_3 of $S{-}step_{11}$; (i) R_3 of $MRF{-}TF_{11}$; (j) R_3 of $SLIC{-}TF_{11}$; (k) R_3 of $FCM{-}TF_{11}$; and (l) R_3 of our proposed STF_{11}.

[68], which exposes that local accumulation of water may lead to a decrease in the performance of large-scale flood extents detection using our STF algorithm. Fig. 13(g1) and (g2) shows that there has been a sharp rising of missed alarms when nonconsecutive suspected flood areas first appear. This kind of error can be corrected by continuously adding new MSH-SITS. When the subsequent flood detection results of the new scenes also cover the newly emerged suspect flood area, our STF-based strategy well reduces missed alarms, such as Fig. 13(h1) and (h2).

Another important factor in achieving better flood detection performance in MSH-SITS data is reducing false alarms caused by cloud and shadow interference. In Fig. 14, we compared the ability of the ten algorithms in Table IV on reducing false alarms under strong cloud shadow interference. Specifically, in Fig. 14, the flood detection results of the ten methods are magnified to show the region (R_3) in Ψ_{11} that has been contaminated by clouds and cloud shadows. It can be seen that our S-step strategy effectively suppresses local false alarms in detecting real flood areas, outperforming the FCM [63], SLIC [64], MRF [25], and Unet++ [61]. Due to the addition of flood contour lines and boundary suppression modules during the training process, XBF [62] algorithm suppressed the interference of cloud shadows and get lower false alarms than FCM [63], SLIC [64], MRF [25], Unet++ [61], and our S-step algorithms. Moreover, Table IV and Figs. 10 and 14 indicate these four methods, i.e., MRF-TF, FCM-TF, SLIC-TF, and our STF, based on our proposed temporal fusion strategy achieve the relatively lower false alarms (caused by

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TABLE V
CD RESULTS OF 12 SCENES BASED ON EIGHT CD ALGORITHMS

Dataset and indicators / Methods	4 homogeneous scenes and 8 heterogeneous scenes		
	OA	KC	F_1
DHFF [56]	0.7212 ±0.0278	0.2959 ±0.0518	0.4108 ±0.0802
HPT [16]	0.8780 ±0.0345	0.4335 ±0.1954	0.4977 ±0.1915
IFBCD [25]	0.8539 ±0.0402	0.4931 ±0.1513	0.5615 ±0.1484
PCANet [57]	0.8837 ±0.0992	0.5571 ±0.2113	0.6102 ±0.1926
S^{3}N [17]	0.8817 ±0.0264	0.4655 ±0.1301	0.5383 ±0.1432
SCCN [19]	0.6821 ±0.1191	0.2597 ±0.1320	0.3780 ±0.1278
Our STFCD with MRF	0.9363 ±0.0204	0.5858 ±0.1887	0.6184 ±0.1915
Our STFCD with SMRF	**0.9727 ±0.0064**	**0.8355 ±0.0824**	**0.8508 ±0.0821**

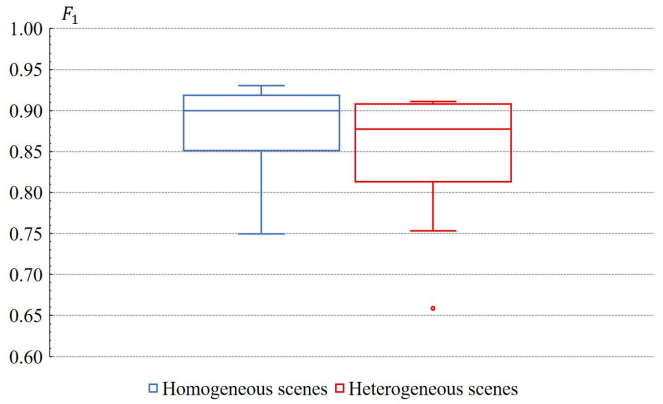


Fig. 15. F_1 boplot of CD results via our STFCD algorithm based on different kinds of scenes.

cloud shadows) by employing the historical flood information, which are visually and numerically superior to their original version without temporal fusion step, i.e., [25], [63], [63] and our S-step in Table IV. In summary, our STF obtains relatively better flood detection performance than the other nine algorithms in Table IV benefiting from the effectively suppression of the interference of cloud shadows and terrain shadows in flood extent detection using the proposed STF algorithm.

C. Flood CD Analysis Based on 12 Scenes in MSH-SITS

In the 12 scenes of HT, we performed CD analysis based on the proposed STFCD algorithm with the comparison of seven CD algorithms, including DHHF [56], HPT [16], IFBCD [25], PCANet [55], S^{3}N [17], SCCN [19], and our STFCD with MRF. The quantitative evaluation results of measured scenes, including four homogeneous scenes (i.e., $\boldsymbol{\Psi_7 - \Psi_{10}}$, $\boldsymbol{\Psi_8 - \Psi_{10}}$, $\boldsymbol{\Psi_8 - \Psi_{12}}$, and $\boldsymbol{\Psi_9 - \Psi_{11}}$) and eight heterogeneous scenes (i.e., $\boldsymbol{\Psi_1 - \Psi_3}$, $\boldsymbol{\Psi_1 - \Psi_5}$, $\boldsymbol{\Psi_1 - \Psi_6}$, $\boldsymbol{\Psi_4 - \Psi_6}$, $\boldsymbol{\Psi_7 - \Psi_9}$, $\boldsymbol{\Psi_7 - \Psi_{11}}$, $\boldsymbol{\Psi_8 - \Psi_{11}}$, and $\boldsymbol{\Psi_9 - \Psi_{12}}$), using eight CD algorithms are listed in Table V. Table V shows that our proposed STFCD-based algorithms, i.e., our STFCD with MRF and our STFCD with SMRF, achieve better CD results than other six algorithms with higher OA, KC, and F_1 values. As shown in Tables III–V, our proposed SMRF outperforms MRF in the STFCD framework on 12 different scenes in terms of numerical results. This is because we adaptively update the local pixel thresholds of suspected flood areas using the SMRF energy function based on the physical characteristics of floods in the infrared band and SAR, which further reduces the missed alarms and false alarms.

Specifically, the OAs, KCs, and F_1s of our STFCD with SMRF have means of 0.9727 ± 0.0064 (standard deviation, s.d.), 0.8355±0.0824 (s.d.), and 0.8508±0.0821 (s.d.), respectively. Based on the statistical results of Table V, we apply a boxplot to evaluate the stability of the flood CD using our STFCD algorithm in the case of homogeneous scenes and heterogeneous scenes in Fig. 15. In Fig. 15, the median line and the box size in homogeneous scenes are higher and smaller than that in heterogeneous scenes, respectively, indicating that the CD results in homogeneous scenes are more accurate and stable than that in heterogeneous scenes due to the influence of illumination and terrain shadows. The emergence of outliers in Fig. 15 denotes the greater challenge of CD in heterogeneous scenes than the homogeneous scenes.

We simultaneously catch sight of original image pairs and the corresponding CD results of each algorithm in Figs. 16–19. The black, red, green, and blue colors represent truly detected nonflood areas, falsely detected flood areas, truly detected flood areas, and falsely detected nonflood areas, respectively. The subimages in Figs. 18 and 19 include screenshots of MSH-SITS, GTs of flood change, and CD results produced by eight algorithms, i.e., DHFF [56], HPT [16], IFBCD [25], PCANet [57], S^{3}N [17], SCCN [19], our STFCD with MRF, and our STFCD with SMRF. Specifically, DHFF can effectively detect most of the change regions in MSH-SITS while introducing high false alarms caused by the mixing of flood changes and other changes. HPT, i.e., a semisupervised algorithm, partly suppresses some false alarms by training the homogeneous transfer feature space, but the incompleteness of training samples (i.e., artificially selected invariant image patches) will cause a lot of missed alarms. Benefiting from the pixel-by-pixel threshold iteration strategy based on the modified MRF, IFBCD has a better suppression capability on false alarms than DHFF, while IFBCD is not stable in the flood CD of heterogeneous scenes. PCANet achieves better flood CD performance than DHFF, HPT, and IFBCD especially in CD of SAR images thanks to the utilization of the representative neighborhood features of each pixel and PCA filters, resulting in the strong inhibition to speckle noises in SAR images. The flood CD performance of S^{3}N in MSH-SITS is more stable than that of PCANet, i.e., the fluctuation range of evaluation indicators for flood CD of the S^{3}N is smaller than PCANet, but S^{3}N still has many missed alarms in the flood CD results due to the incompleteness of artificial selection training samples which is similar to HPT. With manual parameter adjustment, SCCN still has high false alarms in flood CD on MSH-SITS, indicating SCCN is more suitable for small scenes

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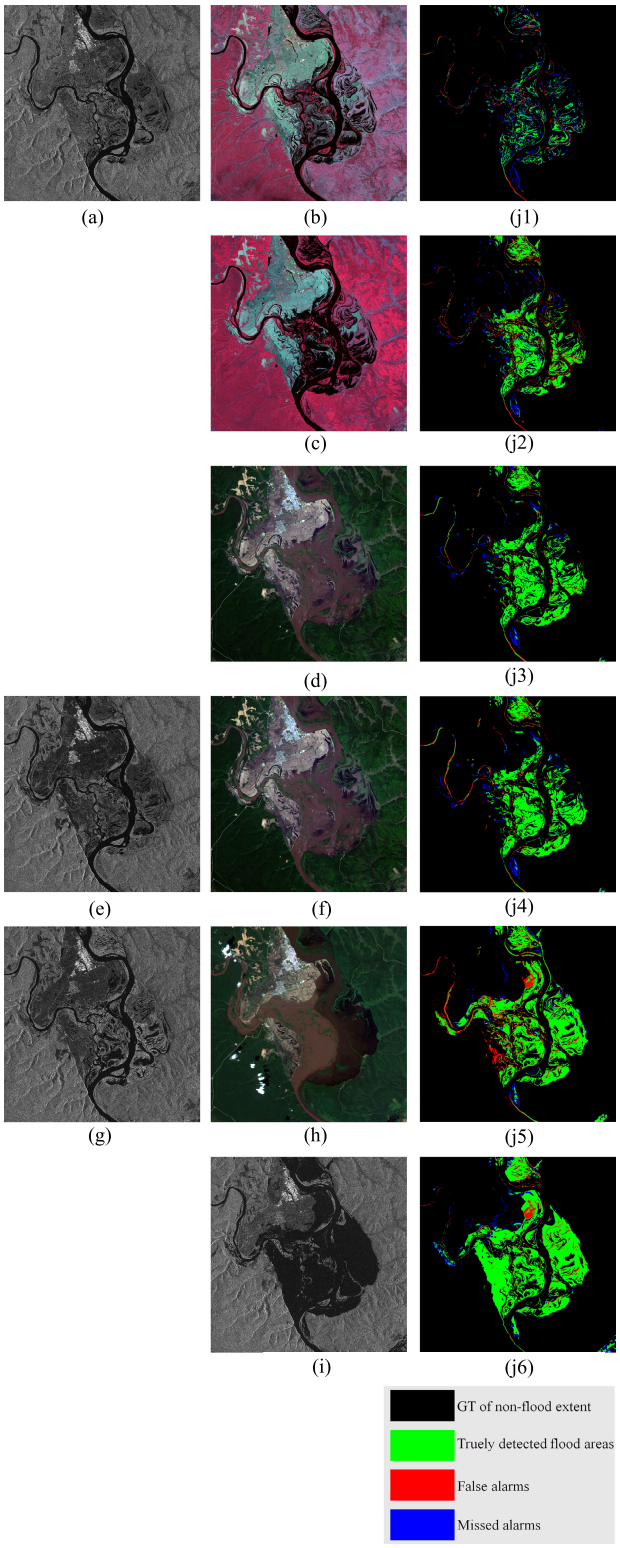


Fig. 16. Screenshots of MSH-SITS and CD results performed by our STF algorithm based on first–sixth scenes: (a), (b), and (j1) thumbnails in the first row are $\boldsymbol{\Psi_1}$, $\boldsymbol{\Psi_3}$, and CD results between $\boldsymbol{\Psi_1}$ and $\boldsymbol{\Psi_3}$ derived from STFCD ($\boldsymbol{STFCD_{1\rightarrow 3}}$), respectively, where "$\mathbf{1 \rightarrow 3}$" indicates the CD operation between $\boldsymbol{\Psi_1}$ and $\boldsymbol{\Psi_3}$; (c) and (j2) screenshots of $\boldsymbol{\Psi_5}$ and $\boldsymbol{STFCD_{1\rightarrow 5}}$ are shown in the second row, respectively; (d) and (j3) downsampling images of $\boldsymbol{\Psi_6}$ and $\boldsymbol{STFCD_{1\rightarrow 6}}$ are demonstrated in the third row, respectively; (e), (f), and (j4) thumbnails of $\boldsymbol{\Psi_4}$, $\boldsymbol{\Psi_6}$, and $\boldsymbol{STFCD_{4\rightarrow 6}}$ are exhibited in the fourth row, respectively; (g), (h), and (j5) contractible images of $\boldsymbol{\Psi_7}$, $\boldsymbol{\Psi_9}$, and $\boldsymbol{STFCD_{7\rightarrow 9}}$ are presented in the fifth row, respectively; and (i) and (j6) shrunken images of $\boldsymbol{\Psi_{10}}$ and $\boldsymbol{STFCD_{7\rightarrow 10}}$ are displayed in the sixth row, respectively.

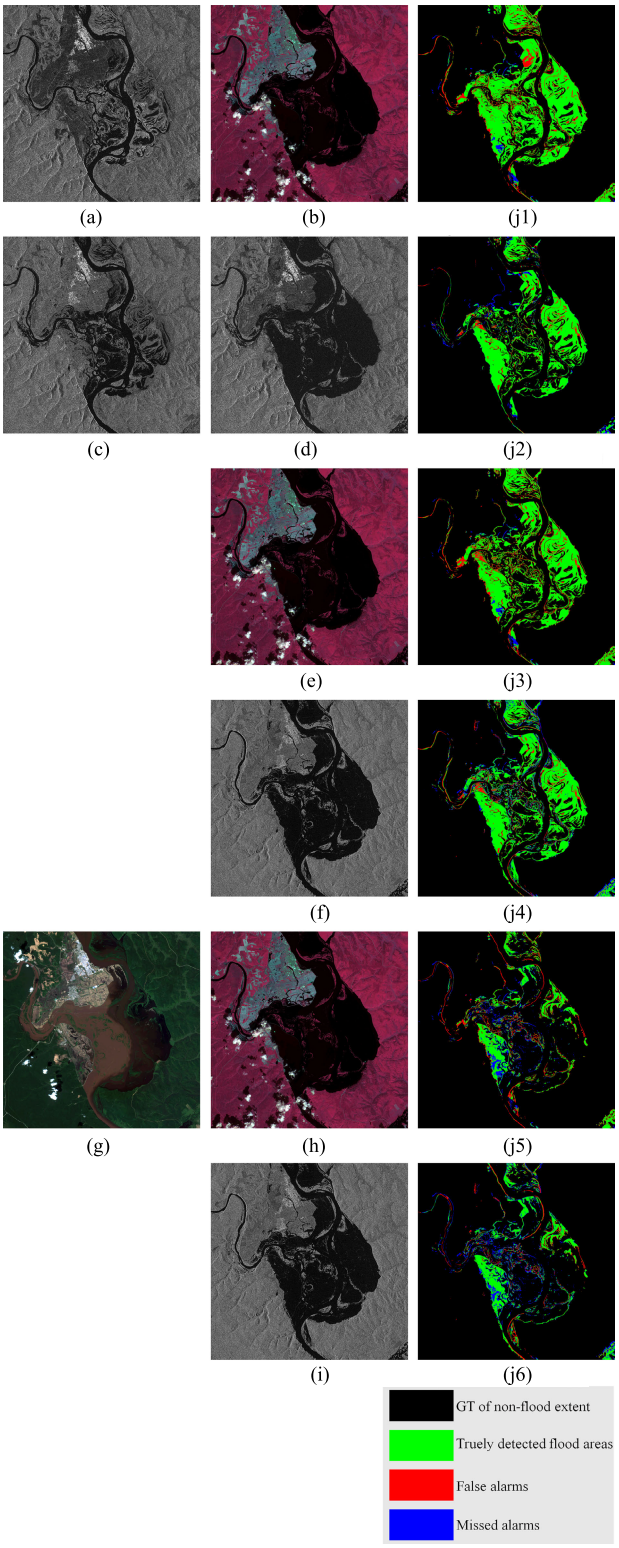


Fig. 17. Thumbnails of MSH-SITS and CD results produced by our STF algorithms originated from seventh–12th scenes: (a), (b), and (j1) shrunken images in the first row are $\boldsymbol{\Psi_7}$, $\boldsymbol{\Psi_{11}}$, and CD results between $\boldsymbol{\Psi_7}$ and $\boldsymbol{\Psi_{11}}$ emanate from STFCD ($\boldsymbol{STFCD_{7\rightarrow 11}}$), respectively, where "$\mathbf{7 \rightarrow 11}$" denotes the CD operation between $\boldsymbol{\Psi_7}$ and $\boldsymbol{\Psi_{11}}$; (c), (d), and (j2) screenshots of $\boldsymbol{\Psi_8}$, $\boldsymbol{\Psi_{10}}$, and $\boldsymbol{STFCD_{8\rightarrow 10}}$ are shown in the second row, respectively; (e) and (j3) downsampling pictures of $\boldsymbol{\Psi_{11}}$ and $\boldsymbol{STFCD_{8\rightarrow 11}}$ are exhibited in the third row, respectively; (f) and (j4) thumbnails of $\boldsymbol{\Psi_{12}}$ and $\boldsymbol{STFCD_{8\rightarrow 12}}$ are demonstrated in the fourth row, respectively; (g), (h), and (j5) contractible images of $\boldsymbol{\Psi_9}$, $\boldsymbol{\Psi_{11}}$, and $\boldsymbol{STFCD_{9\rightarrow 11}}$ are displayed in the fifth row, respectively; and (i) and (j6) thumbnails of $\boldsymbol{\Psi_{12}}$ and $\boldsymbol{STFCD_{9\rightarrow 12}}$ are presented in the sixth row, respectively.

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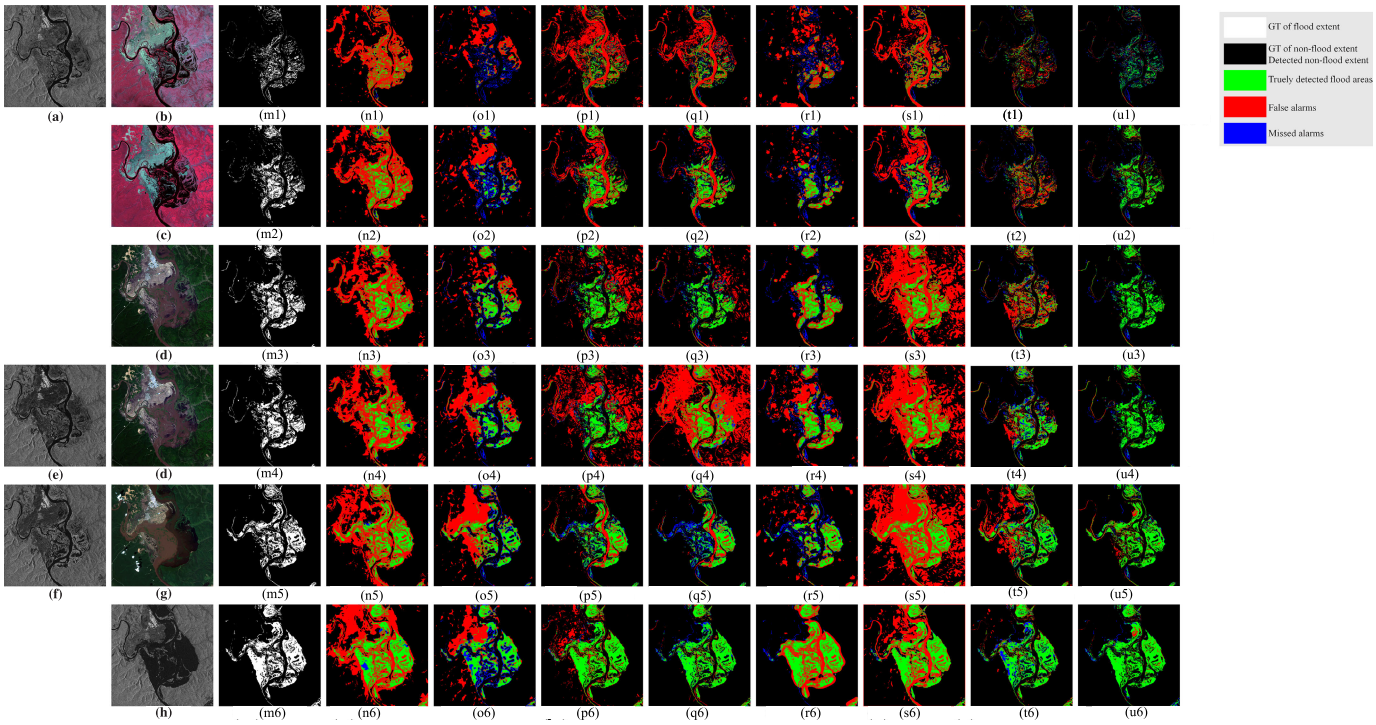


Fig. 18. Screenshots of MSH-SITS, GTs, and CD results performed by eight algorithms based on first–sixth datasets: (a), (b), and (m1)–(u1) thumbnails in the first row are $\boldsymbol{\Psi_1}$, $\boldsymbol{\Psi_3}$, CD results between $\boldsymbol{\Psi_1}$ and $\boldsymbol{\Psi_3}$ derived from GT ($\boldsymbol{GT_{1\to3}}$), DHFF ($\boldsymbol{DHFF_{1\to3}}$), HPT ($\boldsymbol{HPT_{1\to3}}$), IFBCD ($\boldsymbol{IFBCD_{1\to3}}$), PCANet ($\boldsymbol{PCANet_{1\to3}}$), S^{3}N ($\boldsymbol{S^3N_{1\to3}}$), SCCN ($\boldsymbol{SCCN_{1\to3}}$), our proposed STFCD with MRF ($\boldsymbol{STFCD(MRF)_{1\to3}}$), and our proposed STFCD with SMRF ($\boldsymbol{STFCD(SMRF)_{1\to3}}$), respectively, where "$\mathbf{1\to3}$" indicates the CD operation between $\boldsymbol{\Psi_1}$ and $\boldsymbol{\Psi_3}$; (c) and (m2)–(u2) screenshots of $\boldsymbol{\Psi_5}$, $\boldsymbol{GT_{1\to5}}$, $\boldsymbol{DHFF_{1\to5}}$, $\boldsymbol{HPT_{1\to5}}$, $\boldsymbol{IFBCD_{1\to5}}$, $\boldsymbol{PCANet_{1\to5}}$, $\boldsymbol{S^3N_{1\to5}}$, $\boldsymbol{SCCN_{1\to5}}$, $\boldsymbol{STFCD(MRF)_{1\to5}}$, and $\boldsymbol{STFCD(SMRF)_{1\to5}}$ are shown in the second row, respectively; (d) and (m3)–(u3) downsampling images of $\boldsymbol{\Psi_6}$, $\boldsymbol{GT_{1\to6}}$, $\boldsymbol{DHFF_{1\to6}}$, $\boldsymbol{HPT_{1\to6}}$, $\boldsymbol{IFBCD_{1\to6}}$, $\boldsymbol{PCANet_{1\to6}}$, $\boldsymbol{S^3N_{1\to6}}$, $\boldsymbol{SCCN_{1\to6}}$, $\boldsymbol{STFCD(MRF)_{1\to6}}$, and $\boldsymbol{STFCD(SMRF)_{1\to6}}$ are demonstrated in the third row, respectively; (e), (d), and (m4)–(u4) thumbnails of $\boldsymbol{\Psi_4}$, $\boldsymbol{\Psi_6}$, $\boldsymbol{GT_{4\to6}}$, $\boldsymbol{DHFF_{4\to6}}$, $\boldsymbol{HPT_{4\to6}}$, $\boldsymbol{IFBCD_{4\to6}}$, $\boldsymbol{PCANet_{4\to6}}$, $\boldsymbol{S^3N_{4\to6}}$, $\boldsymbol{SCCN_{4\to6}}$, $\boldsymbol{STFCD(MRF)_{4\to6}}$, and $\boldsymbol{STFCD(SMRF)_{4\to6}}$ are exhibited in the fourth row, respectively; (f), (g), and (m5)–(u5) contractible images of $\boldsymbol{\Psi_7}$, $\boldsymbol{\Psi_9}$, $\boldsymbol{GT_{7\to9}}$, $\boldsymbol{DHFF_{7\to9}}$, $\boldsymbol{HPT_{7\to9}}$, $\boldsymbol{IFBCD_{7\to9}}$, $\boldsymbol{PCANet_{7\to9}}$, $\boldsymbol{S^3N_{7\to9}}$, $\boldsymbol{SCCN_{7\to9}}$, $\boldsymbol{STFCD(MRF)_{7\to9}}$, and $\boldsymbol{STFCD(SMRF)_{7\to9}}$ are presented in the fifth row, respectively; (h) and (m6)–(u6) shrunken images of $\boldsymbol{\Psi_{10}}$, $\boldsymbol{GT_{7\to10}}$, $\boldsymbol{DHFF_{7\to10}}$, $\boldsymbol{HPT_{7\to10}}$, $\boldsymbol{IFBCD_{7\to10}}$, $\boldsymbol{PCANet_{7\to10}}$, $\boldsymbol{S^3N_{7\to10}}$, $\boldsymbol{SCCN_{7\to10}}$, $\boldsymbol{STFCD(MRF)_{7\to10}}$, and $\boldsymbol{STFCD(SMRF)_{7\to10}}$ are displayed in the sixth row, respectively (white: GT of incremental flood areas; black: nonflood areas; green: incremental flood areas are correctly detected; red: unchanged areas are incorrectly detected as changed flood areas; and blue: incremental flood areas are incorrectly detected as unchanged areas).

without cloud shadows or terrain shadows. Our STFCD with MRF comprehensively utilizes MRF and our proposed STF strategy and obtains better flood CD results in MSH-SITS than [16], [17], [19], [25], [56], and [57].

As shown in Figs. 16–19, there is a significant difference between our STFCD with SMRF and the other seven comparison algorithms, where our STFCD with SMRF always remains the better flood CD results with fewer false alarms and missed alarms. Furthermore, Figs. 16–19 show that all eight algorithms obtain good CD results in the homogenous SAR scenes (i.e., $\boldsymbol{\Psi}_1-\boldsymbol{\Psi}_{10}$ and $\boldsymbol{\Psi}_8-\boldsymbol{\Psi}_{10}$). It can be seen that most CD algorithms suffer from the dramatic changes of the data distribution especially in the heterogeneous scenes, resulting in the degradation of flood CD performance. Our STFCD algorithm exhibits relatively better domain generalization ability in the flood CD tasks, benefiting from the comprehensive exploitation of both the spatial characteristics of a single image in MSH-SITS and the temporal correlation in MSH-SITS.

V. Conclusion

In this study, we innovatively proposed an STFCD algorithm for the continuous monitoring of flood extents using MSH-SITS. The flood monitoring framework of our proposed STFCD algorithm mainly contains two main steps, i.e., spatial clustering and temporal fusion. In the spatial clustering step, we propose the SMRF algorithm to adaptively extract the main and partial flood features, benefiting from the spatial dependence of MSH-SITS. Next, in the temporal fusion step, we develop a temporal fusion strategy based on the historical flood information to exploit the complementary characteristics in MSH-SITS, which further optimize the flood CD results. Compared with existing methods, our main contribution is to better utilize the correlation characteristics of local neighborhood space using the proposed SMRF algorithm with the adaptive receptive fields based on the single-source images and efficiently exploit the temporal interaction of flood features in MSH-SITS to enhance the reliability of the CD results. The proposed STFCD algorithm was successfully applied in the flood extent monitoring in HT. Our proposed STFCD algorithm maintains dominant performance in continuous flood CD in all 12 scenes of the MSH-SITS in HT. Moreover, the generation of main flood maps by our STF-based strategy also leads to excellent consistency with the actual flood occurrence.

In the future, we will collect more MSH-SITS in worldwide and try to reduce the false flood detection during or after the rainfall.

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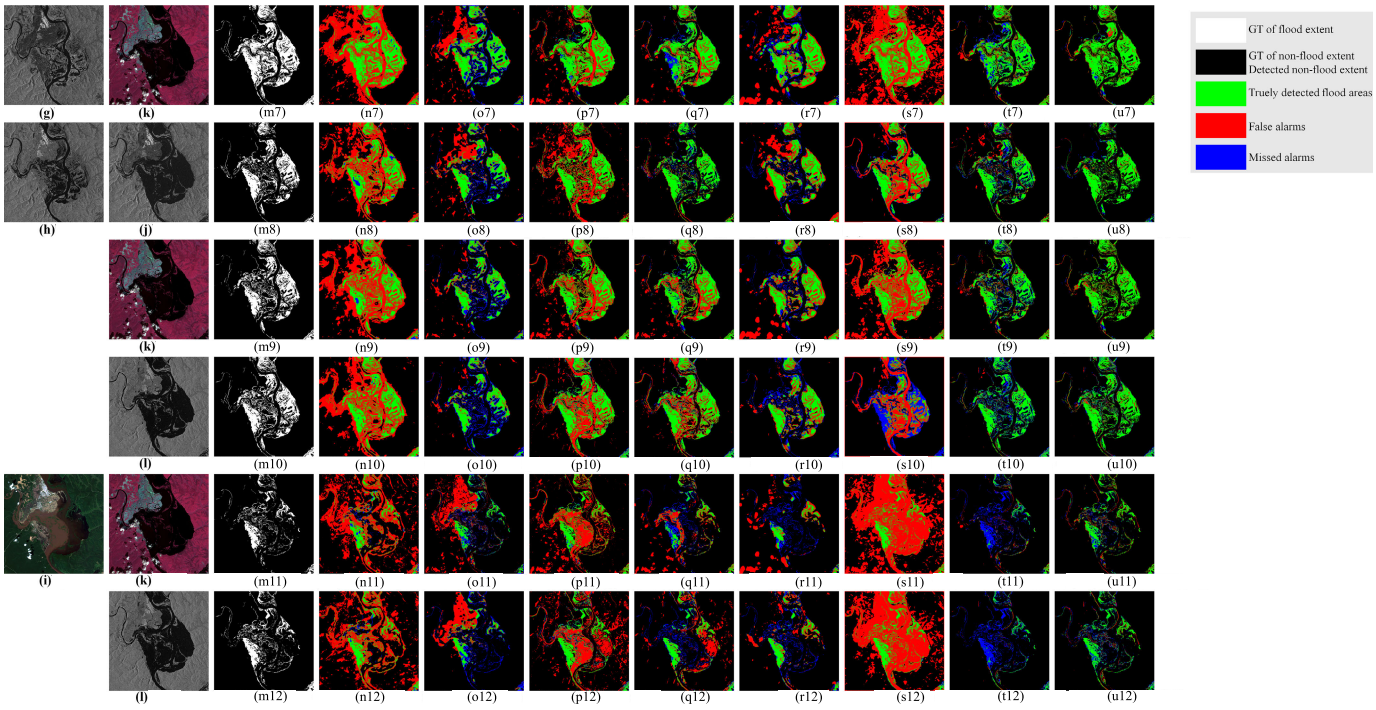


Fig. 19. Thumbnails of MSH-SITS, GTs, and CD results produced by eight algorithms originated from seventh–12th datasets: (g), (k), and (m7)–(u7) shrunken images in the first row are Ψ_7, Ψ_{11}, CD results between Ψ_7 and Ψ_{11} emanate from GT ($GT_{7\to11}$), DHFF ($DHFF_{7\to11}$), HPT ($HPT_{7\to11}$), IFBCD ($IFBCD_{7\to11}$), PCANet ($PCANet_{7\to11}$), S^{3}N ($S^3N_{7\to11}$), SCCN ($SCCN_{7\to11}$), our proposed STFCD with MRF ($STFCD(MRF)_{7\to11}$), and our proposed STFCD with SMRF ($STFCD(SMRF)_{7\to11}$), respectively, where "$\mathbf{7\to11}$" denotes the CD operation between Ψ_7 and Ψ_{11}; (h), (j), and (m8)–(u8) screenshots of Ψ_8, Ψ_{10}, $GT_{8\to10}$, $DHFF_{8\to10}$, $HPT_{8\to10}$, $IFBCD_{8\to10}$, $PCANet_{8\to10}$, $S^3N_{8\to10}$, $SCCN_{8\to10}$, $STFCD(MRF)_{8\to10}$, and $STFCD(SMRF)_{8\to10}$ are shown in the second row, respectively; (k) and (m9)–(u9) downsampling pictures of Ψ_{11}, $GT_{8\to11}$, $DHFF_{8\to11}$, $HPT_{8\to11}$, $IFBCD_{8\to11}$, $PCANet_{8\to11}$, $S^3N_{8\to11}$, $SCCN_{8\to11}$, $STFCD(MRF)_{8\to11}$, and $STFCD(SMRF)_{8\to11}$ are exhibited in the third row, respectively; (l) and (m10)–(u10) thumbnails of Ψ_{12}, $GT_{8\to12}$, $DHFF_{8\to12}$, $HPT_{8\to12}$, $IFBCD_{8\to12}$, $PCANet_{8\to12}$, $S^3N_{8\to12}$, $SCCN_{8\to12}$, $STFCD(MRF)_{8\to12}$, and $STFCD(SMRF)_{8\to12}$ are demonstrated in the fourth row, respectively; (i), (k), and (m11)–(u11) contractible images of Ψ_9, Ψ_{11}, $GT_{9\to11}$, $DHFF_{9\to11}$, $HPT_{9\to11}$, $IFBCD_{9\to11}$, $PCANet_{9\to11}$, $S^3N_{9\to11}$, $SCCN_{9\to11}$, $STFCD(MRF)_{9\to11}$, and $STFCD(SMRF)_{9\to11}$ are displayed in the fifth row, respectively; (l) and (m12)–(u12) thumbnails of Ψ_{12}, $GT_{9\to12}$, $DHFF_{9\to12}$, $HPT_{9\to12}$, $IFBCD_{9\to12}$, $PCANet_{9\to12}$, $S^3N_{9\to12}$, $SCCN_{9\to12}$, $STFCD(MRF)_{9\to12}$, and $STFCD(MRF)_{9\to12}$ are presented in the sixth row, respectively (white: incremental flood regions; black: nonchanged flood regions; green: true positive; red: false alarms; and blue: missed alarms).

APPENDIX

See Figs. 18 and 19.

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Zhihao Wang (Student Member, IEEE) received the B.S. degree from the Heilongjiang University of Science and Technology, Harbin, China, in 2018, and the M.S. degree in electronic engineering from Tsinghua University, Beijing, China, in 2021, where he is currently pursuing the Ph.D. degree.

His main research interests include remote sensing image processing, change detection, and multisource remote sensing data fusion.



Xueqian Wang (Member, IEEE) received the B.S. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2015, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2020.

From 2018 to 2019, he visited Syracuse University, Syracuse, NY, USA. He is currently an Assistant Professor with the Department of Electronic Engineering, Tsinghua University. His main research interests include target detection, information fusion, radar imaging, compressed sensing, and distributed signal processing.



Wei Wu received the Ph.D. degree in geography from Peking University, Beijing, China, in 2017.

Since 2006, he has been a Staff of the National Disaster Reduction Center of China, Beijing, where he is currently the Director and a Researcher with the Disaster Assessment Department. He has authored more than 30 academic articles, holds four national invention patents, and presided over and participated in the preparation of three national standards. His research field is natural disaster monitoring and evaluation. In recent years, he has been mainly engaged in the research and application of natural disasters monitoring, disaster risk assessment, disaster emergency response and comprehensive loss assessment, and disaster reduction application product development and services.



Gang Li (Senior Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from Tsinghua University, Beijing, China, in 2002 and 2007, respectively.

Since July 2007, he has been a Faculty Member with Tsinghua University, where he is currently a Professor with the Department of Electronic Engineering. From 2012 to 2014, he visited The Ohio State University, Columbus, OH, USA, and Syracuse University, Syracuse, NY, USA. He has authored or coauthored more than 200 journal articles and conference papers. He has authored the book *Advanced Sparsity-Driven Models and Methods for Radar Applications* (London, U.K.: SciTech/IET Publishing, in 2020) and edited the book *Radar Countermeasures for Unmanned Aerial Vehicles* (SciTech/IET Publishing, in 2021). His research interests include radar signal processing, distributed signal processing, remote sensing, and information fusion.

Prof. Li is a Senior Area Editor of the IEEE SIGNAL PROCESSING LETTERS and an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. He was also an Associate Editor of the IEEE TRANSACTIONS ON SIGNAL PROCESSING, a Special Sessions Chair for the International Conference on Acoustics, Speech, and Signal Processing 2021, and a Tutorial Speaker for the IEEE Radar Conference 2017.

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